

Obstetric Medicine

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Friday, November 28, 2025



List of Short Forms in This Talk

SF	SHORT FORM
ACOG	American College of OB/Gyne (acog.org)
APLA	Antiphospholipid antibodies
BG	BLOOD GLUCOSE
CHD	CONGENITAL HEART DISEASE
HDP	Hypertensive disorders of Pregnancy
FM	FETAL MALFORMATIONS (Baseline rate 1-3% in healthy pregnancies)
HELLP	HEMOLYSIS, ELEVATED LIVER ENZYMES, LOW PLATELETS
ICS, SABA	INHALED CORTICOSTEROID
PET	PRE ECLAMPSIA / TOXEMIA (we don't use word toxemia anymore though)
PTL	PRE TERM LABOUR
SGA (LGA)	Small for Gestational Age (Large for GA)
SOGC	Society of OB/Gyne Canada (www.sogc.org)
UPCr	Urine Protein:Creatinine Ratio
VTE	VENOUS THROMBO EMBOLISM (DVT OR PE)

Outline

- **Hypertension & Pre-eclampsia**
- VTE and ITP in pregnancy
- Diabetes in pregnancy
- High Yield GI disorders of pregnancy
- High Yield Cardio in pregnancy
- High Yield Resp in pregnancy
- **BONUS SLIDES**
 - High Yield Infections in pregnancy
 - High Yield Heme disorders of pregnancy
 - Drugs in Pregnancy
 - Approach to the Obstetric Patient in a Scenario
 - Pre-pregnancy counselling for women with chronic conditions
 - Normal Labs in Pregnancy
- **COVERED ELSEWHERE:**
 - Thyroid → ENDO
 - Asthma Treatment → RESP
 - HIV and others in pregnancy → ID

Hypertensive Disorders of Pregnancy

- Key Resources:
 - **SOGC 2022 Guideline:** Magee LA, et al. J Obstet Gynaecol Can. 2022 May;44(5):547-571.
 - **Hypertension Canada 2020 Guidelines:** Rabi DM et al. Can J Cardiol 36(2020)- 596-624.
 - **SOGC 2014 Guideline:** Magee LA et al. J Obst Gynae Canada 2014; 36(5):416-438.
 - Control of Hypertension in Pregnancy (**CHIPS**) Trial (Magee LA et al, NEJM 2015; 372(5)).
 - **CHAP:** Magee LA, von Dadelszen P. Treatment for Mild Chronic Hypertension during Pregnancy. N Engl J Med. 2022 Aug 18;387(7):664. doi: 10.1056/NEJMc2207889. PMID: 36070722.
 - Magnesium sulfate for prevention of eclampsia in pre-eclampsia (**Magpie**) Trial (Altman et al, Lancet 2002; 359(9321)).
 - Aspirin versus Placebo in Pregnancies at High Risk for Preterm Preeclampsia (**ASPREE**) Trial (Rolnik et al, NEJM 2017; 377(7): 613-622).

MCQ 1 2026

42F G1P0 currently 28 weeks gestation conceived via IVF presents to the hospital after a seizure. She has no prior history of seizures and a history of lupus arthritis in remission on Prednisone 10mg daily and Hydroxychloroquine. She has been on ASA 162mg daily since 10 weeks. She had 2 minutes of full body convulsion with tongue biting followed by 20 minutes of confusion. On exam, BP 170/110, HR 110, O2 sat 96% on RA. She is lethargic, but alert and oriented without focal neurologic symptoms. Blood work shows Hb of 100, Plt of 45, AST 560, ALT 640, ALP 150, Bilirubin of 10, Cr of 90, PLGF of 25, and urine PCR of 340. Ultrasound shows a fetal growth of 3rd percentile with absent end diastolic flow of the uterine artery. CT head is nil acute.

Which of the following is not the next step?

- a) Labetalol 10mg IV x 1 dose then Labetalol 200mg po TID
- b) MgSO4 4g IV x1 then 1g/hr infusion x 24 hours
- c) Phenytoin 20mg/kg IV x 1 then 300mg po daily
- d) Betamethasone 12mg IM x 1
- e) Urgent delivery
- f) Investigation for antiphospholipid syndrome postpartum

Classification

- Pre-existing hypertension



Prior to 20 weeks of gestation

- Gestational hypertension



After 20 weeks of gestation

- Pre-eclampsia/eclampsia



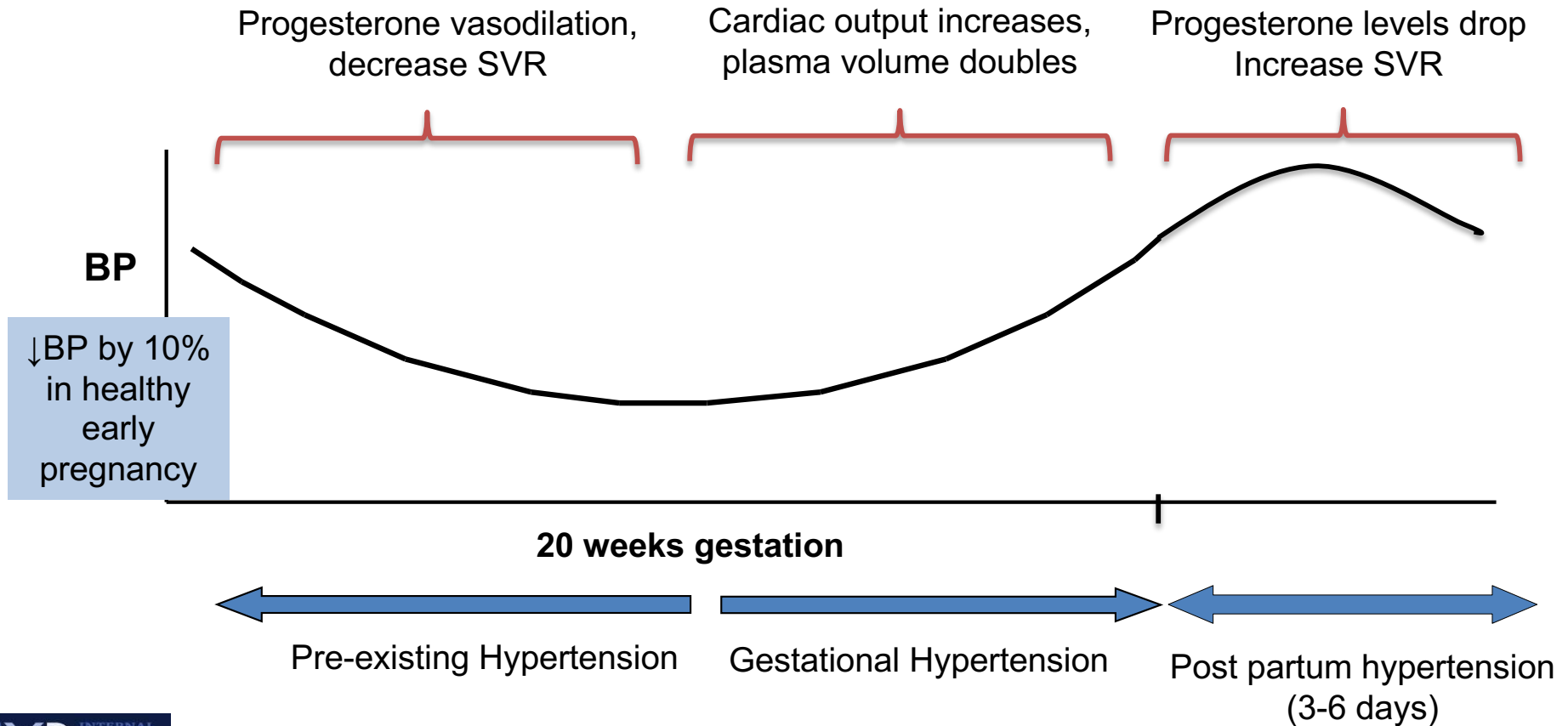
New or worsening hypertension, with one or more adverse conditions (usually after 20 weeks)

- Others

- White coat hypertension
- Masked hypertension
- Transient hypertension
- Resistant hypertension

SOGC 2022

Physiologic changes in pregnancy



Diagnostic Criteria

Hypertension

Office:

SBP $\geq$ 140 or DBP $\geq$ 90mmHg

Ambulatory BP monitor:

SBP $\geq$ 135 or DBP $\geq$ 85mmHg

*Average of 2
measurements,
15 minutes
apart

Severe
hypertension

SBP $\geq$ 160 or DBP $\geq$ 110 mmHg



Go to ER!
↑risk of ICH

Resistant
hypertension

$\geq$ 3 anti-hypertensives

SOGC 2022

BP Target

Hypertension Canada 2020
SOGC 2022

DBP < 85 mmHg

CHIPS (NEJM 2015)

***Patients with
pregestational diabetes**
should have a BP target
of **<130/80** throughout
pregnancy

- RCT of 987 women
- Tight (DBP<85) vs less tight (DBP <100)
- No difference in fetal or maternal outcomes between groups.
- Higher severe hypertension 160/110 in less tight group (40% vs 27.5%), and had more lab features of HELLP

SBP target in practice

SBP 130-140

Treatment

Non-severe hypertension

Drug	Starting dose	Max dose	Notes
Labetalol	100mg BID-TID	400mg TID or 300mg QID	Avoid in bradycardia or severe asthma
Methyldopa	250mg TID	750mg TID (2.5g/daily)	Fatigue “methyldopey”
Nifedipine XL	30mg daily	60mg BID(120mg/day)	Not to be confused with IR
HCTZ	12.5mg daily	50mg daily	<i>Second line – prefer meds above!</i>



Severe hypertension (≥ 160 systolic or ≥ 110 diastolic) – lower within 30-60 min

Drug	Dose
Nifedipine IR	5-10mg po*, repeat q30min, *chew for rapid onset
Labetalol	10-20mg IV push, repeat q30min
Hydralazine	5-10mg IV, repeat q30min

Breastfeeding: All of the above, plus selected ACEi (enalapril, captopril, quinapril)



Drugs to avoid in pregnancy!

Atenolol: IUGR in pregnancy, fetal bradycardia in breast feeding

ACEi/ARB: fetal renal agenesis, oligohydramnios, pulmonary aplasia, especially in T2/T3

Prazosin – uncertain data, avoid esp peripartum

Benefits of BP control*

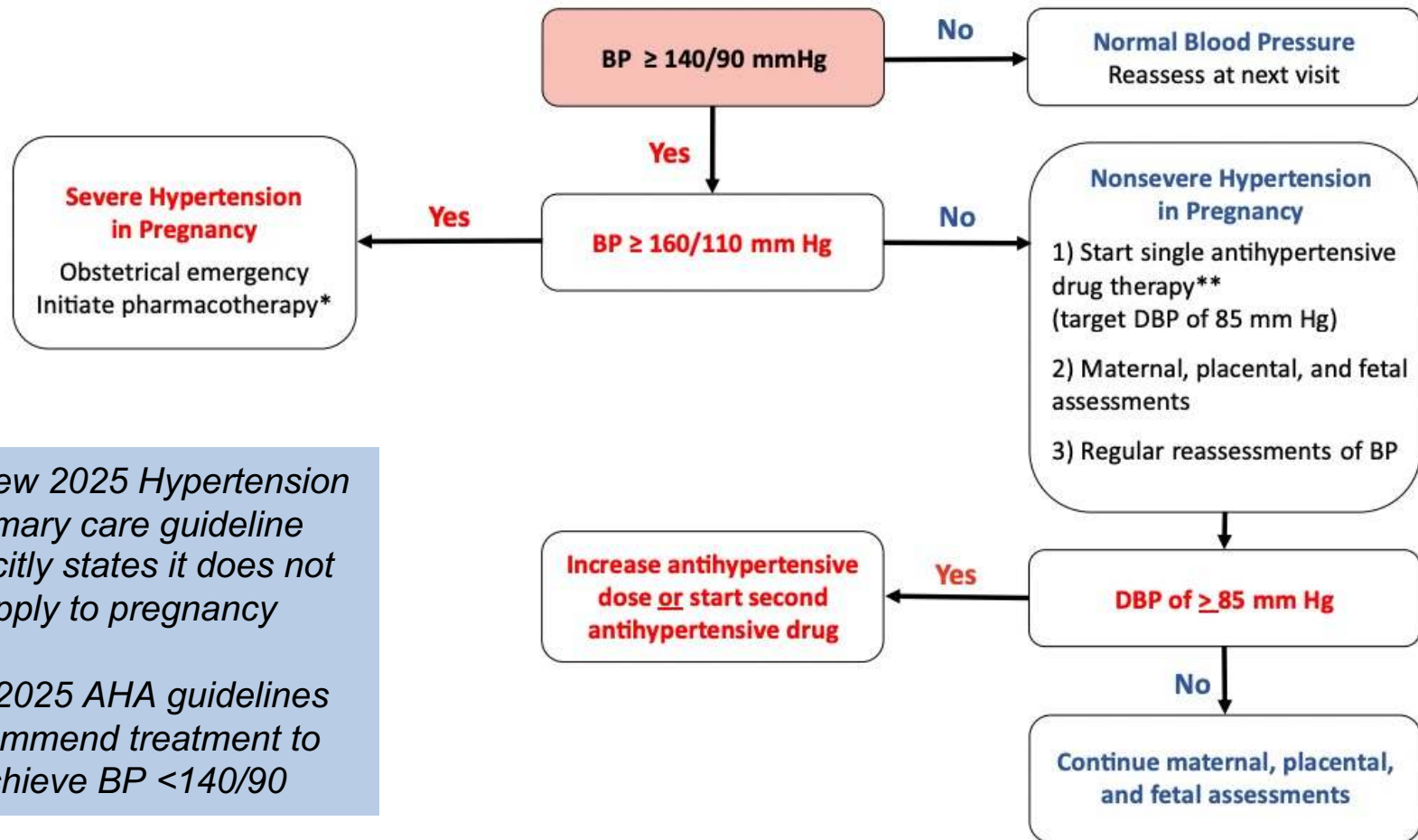
- Treatment of non-severe hypertension Reduces risk of severe hypertension¹
- Treating to BP <140/90 (vs 160/105) reduces composite endpoint (preeclampsia, preterm birth, placental abruption, fetal / neonatal death) ²
- Significantly reduced pre-eclampsia with severe features, medically indicated preterm births <35 weeks, placental abruption, fetal or neonatal death. Safe for baby as no difference in small for gestational age <10th percentile.³

1Cochrane Database Syst Rev. 2018;10:CD002252

2Obstet Gynecol. 2024 Jul 1;144(1):126-134.

3NEJM 2022 386(19):1781–1792

Hypertension 2020 Treatment algorithm



The new 2025 Hypertension Primary care guideline explicitly states it does not apply to pregnancy

The 2025 AHA guidelines recommend treatment to achieve BP $<140/90$

Preeclampsia

- Definition
 - Multisystem disorder in pregnancy, **placental mediated**, resulting in systemic vascular and endothelial dysfunction, resulting in maternal and fetal morbidity and mortality

• Risk factors	Epidemiologic	■ New partner ■ Family history (mom, sister) ■ Age < 20 or > 35
	Maternal HIGH RISK FACTORS	■ Previous pre-eclampsia ■ Chronic hypertension ■ Diabetes ■ Renal disease ■ Auto-immune disorders (SLE, APLA) ■ BMI>30 ■ Assist reproductive therapy (IVF)
	Maternal MODERATE RISK FACTORS	■ Age > 40 ■ Nulliparity or multiple gestation ■ Prior abruption, stillbirth or fetal growth restriction

Give ASA for prevention if:

- ANY high risk
- 2+ moderate risk factors

Angiogenic Growth Factors in Pregnancy

- SOGC 2022 for the first time has recommended where available assessment of specific biomarkers of preeclampsia:
 - **sFLT-1** [inflammatory biomarker, ↑ with preeclampsia]
 - Placental Growth Factor (**PLGF**) [↓ with placental insufficiency]
 - Ratio of the two (sFlt : PLGF) [cutoff depends on assay used]
- Preeclampsia prediction models incorporating clinical factors, biomarkers and fetal placental ultrasound parameters (e.g. uterine artery resistance and flow) are recommended for use
 - FYI risk calculator available now www.fetalmedicine.org → calculators

Not expected to be testable in GIM exam.
Just FYI as will see reference to these levels in clinical practice at high-risk OB centre!

Preeclampsia Diagnosis

Hypertension

- BP >140/90 x 2 or >160/110 x 1

Plus one of the following three criteria:

1) new or worsening proteinuria

- $\geq 2+$ on dipstick, urine PCR >30mg/mmol, urine ACR >8mg/mmol, or >300mg of protein on 24-hour urine

2) one or more adverse conditions

- **Maternal symptoms:** headache/vision changes, dyspnea/chest pain, RUQ pain, new onset edema/anasarca, seizures, clonus
- **Lab abnormalities:** Hb, Plt, AST/ALT/ALP, Cr, ~~uric acid~~, LDH, bilirubin, blood film, fibrinogen, INR/PTT
- **Fetal complications:** IUGR, oligohydramnios, absent/reversed end diastolic flow on placental doppler

3) one or more complications

***HELLP:** Hemolysis, Elevated Liver enzymes, Low Platelets, is a complication of preeclampsia

~~Uric acid~~ no longer recommended in routine labs by SOGC 2022

SOGC 2022

Organ System	Follow closely regarding need for delivery	Delivery regardless of gestational age (Severe preeclampsia, warrants MgSO ₄)
CNS	Headache / visual symptoms	Eclampsia, PRES, Cortical blindness GCS <13 Stroke, TIA or Reversible ischemic neuro deficit
Cardio-resp	Chest pain / dyspnea O ₂ <97%	Uncontrolled severe HTN (>160/110 over 12h despite 3 anti-hypertensives) O ₂ <90%, need for >50% O ₂ for 1 hr, intubation, pulmonary edema, inotropic support, MI
Heme	Low Plt <100	Plt <50 Transfusion of any blood product
Renal	Elevated serum creatinine	AKI (Cr >150, no prior renal disease) New indication for dialysis.
Hepatic	RUQ or epigastric pain Elevated AST, ALT	Liver dysfunction (INR>2 without DIC or warfarin) Hepatic hematoma or rupture
Feto-placental	Abnormal NST IUGR Oligohydramnios Absent/reversed end diastolic flow Angiogenic imbalance*	Abruption Absent or reverse ductus venosus A wave Intrauterine fetal death

This table is reproduced from Table 4 of
SOGC 2022 guideline

Preeclampsia Treatment

1. Hypertension treatment

- BP treatment **DOES NOT** prevent progression of pre-eclampsia
- BP treatment DOES reduce risk of severe hypertension
 - For inpatients with preeclampsia, this is associated with a reduction in stroke risk!

2. Eclampsia prevention

- Magnesium Sulfate

3. Delivery

- Delivery of the placenta provides “source control” to begin resolution of pre-eclampsia

Due to persistence of angiogenic factors post-delivery, BP rises postpartum and up to 25% of women with hypertensive disorders of pregnancy have postpartum HTN – BP check needed within 3-10d post delivery

Magnesium Sulfate

- Magpie trial (Lancet 2002)
 - 10,000 women with pre-eclampsia, randomized to Mg vs placebo
 - 1° outcome eclampsia risk ↓ (RRR 58%, ARR 1.1%, NNT 91), trend towards ↓ maternal mortality
- For **prevention and treatment** of eclampsia (**choose on MCQ!**)
 - Indications for preventive use
 - severe pre-eclampsia, severe hypertension, headache/vision changes/RUQ pain, platelets <100, elevated liver enzymes, renal insufficiency (I-C)
 - First line treatment for seizures in pregnancy related to eclampsia

Dose: 4g IV loading bolus then 1g/hr x 24h
Q30min vitals; Q1h monitor: U/O, reflexes

Caution if oliguric renal failure → Mg Toxicity. Can withhold bolus and reduce infusion rate refer to your hospital protocol or consult with pharmacist.

NO IV ACCESS? MgSO₄ 5g IM into EACH BUTTOCK (10g) load → 5g IM q4h maintenance

Magnesium Toxicity

Symptoms: ↓ BP ↓ HR ↓ GCS,
respiratory suppression



Monitoring: clinically (decreased reflexes, decreased urine output), NOT serum Mg levels

Treatment: stop Mg, give **Calcium Gluconate**, dialysis

Delivery: Decision in conjunction with OB!

"I will admit patient to hospital, consult OB to assess for fetal complications and assess for delivery. I will also consult anesthesia and pediatrics"

- Indications
 - Immediate delivery if severe / complications of pre-eclampsia: HTN maxed on ≥ 3 agents, end organ damage, stroke/MI
 - Fetal complications: IUGR, Doppler abnormalities, fetal distress
- Timing of Delivery:
 - Chronic Hypertension without preeclampsia – delivery can be induced **38-39⁶ weeks**
 - Gestational hypertension – onset <37 weeks, no preeclampsia: consider **38-39⁶ weeks**
 - Preeclampsia: individualize, expectant to 34 weeks, but no later than **37 weeks gestation**
- Antenatal corticosteroids (**Betamethasone** 12mg IM x 2) for fetus <35 wks if delivery anticipated within next 7 days (I-A)
- Total fluid intake in labour <80ml/h based upon risk of pulm edema [SOGC 2022]
- Pre-eclampsia is not a contraindication for **vaginal delivery**
- **Epidural** safe if platelets >75* (ask your anesthesiologist re: cutoffs) & without coagulopathy

Postpartum Care

- BP rises 3-6 days post-partum
 - **SOGC 2022** : “blood pressure should be measured regularly in the first 2 weeks after delivery in women with hypertension” / ACOG: Check BP within day 3-10 postpartum.
 - BP treatment – can include ACE inhibitors are safe in breast feeding (Enalapril, Quinapril, Captopril)

Cardiovascular Risk Counseling

- **HDP = Future health risks** (Johnston et al., *Circulation*, 2025):
 - 4× risk of chronic hypertension
 - 2× risk of heart disease
 - 4× risk of stroke
 - Increased risk of **diabetes, VTE, and atrial fibrillation**
- **Future pregnancy risk – risk of preeclampsia, role of ASA for prevention**
- **AHA 2024**: All patients with adverse pregnancy outcomes should receive:
 - **BP + lipid screening**
 - **Lifestyle counseling** (diet, exercise, smoking cessation)
- **CCS 2021 Lipid Guidelines**:
 - Recommend **lipid profile postpartum** in all patients with hypertensive disorders of pregnancy

VTE Prophylaxis postpartum

- **Consider thromboprophylaxis** if risk factors (e.g., IUGR, C-section)

Pre-eclampsia Prevention

1) ASA - 81-162mg daily (ASPREE NEJM 2017: N798, Aspirin 150mg vs placebo reduced pre-eclampsia <37 weeks (1.6% vs 4.3%, ARR 2.7%, NNT 37)



Start **<16 weeks** (usually 10-12 weeks) continue **until 36 weeks**

- **Indications** (US task force): previous pre-eclampsia, chronic hypertension, type 1 or 2 diabetes, chronic kidney disease, SLE, APLA syndrome, multiple gestation, two or more minor factors (ex. nulliparous or new partner, IVF, Age >35, BMI>30, first degree family history of pre-eclampsia...)

2) Calcium: Recommended to supplement if low dietary intake <900mg dietary calcium (SOGC 2022)

3) Exercise (II-B) - 150 min / week moderate intensity exercise may reduce risk of hypertensive disorders of pregnancy including preeclampsia

- Recommendation requires *controlled HTN*, & has some obstetric contraindications ex placenta previa

4) Monitoring (OB): Placental ultrasound (uterine and umbilical artery doppler) and biomarker testing (PLGF)

Specific Scenarios:

- **OSA.** - Treating OSA with CPAP shown to reduce gestational hypertension and pre-eclampsia in meta-analysis of 800 patients (Lee YC et al., JAMA 2024)
- **APLA / APS and Preeclampsia Prevention (see next slide)**

Obstetric APS

new criteria- ACR/EULAR Sept 2023

Diagnostic criteria

- 1) ≥ 3 consecutive unexplained prefetal deaths $<10^0$ weeks and/or early fetal deaths $> 10^0$ weeks - 15^6 weeks, or
- 2) Fetal death 16^0 weeks - 34^0 weeks, or
- 3) Pre-eclampsia with severe features and/or placental insufficiency* with severe features ($<34^0$ weeks) with or without fetal death



Lupus anticoagulant, anti-cardiolipin, or beta-2-glycoprotein, if positive repeat in 12 weeks

*If obstetric APS treat with **ASA 81mg + prophylactic dose LMWH** in pregnancy and 6 weeks postpartum*

*Placental insufficiency criteria **beyond GIM scope but FYI:**

“fetal/newborn growth restriction $<10^{\text{th}}$ percentile AND at least 1 of”

- Abnormal fetal surveillance (eg non stress test non reactive)
 - Abnormal umbilical artery doppler
 - Severe IUGR $<3^{\text{rd}}$ centile
 - Oligohydramnios
 - Placental pathology showing vascular malperfusion (eg placental infarct)
- NOT placental abruption (not seen w APLA)

NEW criteria for APLAS testing in 2023
NOT widely adopted in Canada – SOGC/MFM
order ‘case by case’

MCQ 1 2025

42F G1P0 currently 28 weeks gestation conceived via IVF presents to the hospital after a seizure. She has no prior history of seizures and a history of lupus arthritis in remission on Prednisone 10mg daily and Hydroxychloroquine. She has been on Aspirin 160mg daily since 12 weeks gestation. She had a tonic-clonic convulsion with tongue biting followed by 20 minutes of postictal lethargy. She is lethargic, but alert and oriented. Vitals: BP 160/110, HR 110, RR 18, SpO2 96% on RA. She is lethargic, but alert and oriented. Hb of 100, Plt of 45, AST 560, ALT 640, ALP 150, Etc. Ultrasound shows a fetal growth of 3rd percentile. Head is nil acute.

Which of the following is not the next step?

- a) Labetalol 10mg IV x 1 dose then Labetalol 10mg PO q6h
- b) MgSO₄ 4g IV x1 then 1g/hr infusion x 24 h
- c) **Phenytoin 20mg/kg IV x 1 then 300mg po q12h**
- d) Betamethasone 12mg IM x 1
- e) Urgent delivery
- f) Investigation for antiphospholipid syndrome

Early pre-eclampsia with HELLP and eclampsia

- a. IV Labetalol to acutely treat severe hypertension and oral Labetalol to maintain normotension
- b. Magnesium sulfate to prevent recurrent seizures with dose adjusted for renal function
- c. **Phenytoin not indicated– the treatment for eclampsia is not with antiepileptics but rather Magnesium and delivery**
- d. Given the fetus is <34 weeks, betamethasone 12mg IM every 24 hours x 2 doses should be given for lung maturation, in this case give one-time dose as delivery is imminent (don't forget to assess for stress dose steroids for your pt on chronic Pred postpartum)
- e) Pre-eclampsia complicated by eclampsia and HELLP is an indication for urgent delivery
- f) Investigate for antiphospholipid syndrome as a cause for early pre-eclampsia should be done 12 weeks postpartum, especially in the context of history of lupus

Outline

- Hypertension & Pre-eclampsia
- **VTE and ITP in pregnancy**
- Diabetes in pregnancy
- High Yield GI disorders of pregnancy
- High Yield Cardio in pregnancy
- High Yield Resp in pregnancy

- BONUS SLIDES
 - High Yield Infections in pregnancy
 - High Yield Heme disorders of pregnancy
 - Drugs in Pregnancy
 - Approach to the Obstetric Patient in a Scenario
 - Pre-pregnancy counselling for women with chronic conditions
 - Normal Labs in Pregnancy
- COVERED ELSEWHERE:
 - Thyroid → ENDO
 - Asthma Treatment → RESP
 - HIV and others in pregnancy → ID

BONUS

Read on own

VTE in Pregnancy

- Key Resource:
 - **Thrombosis Canada** Guidelines, **2021**.
 - **ASH** Guidelines for management of VTE in pregnancy (blood adv. **2018**; 2(22):3317-59).
 - **SOGC** Clinical Practice Guideline (J Obstet Gynaecol Can **2014**; 36(6):527-553)
 - Pregnancy-Adapted YEARS Algorithm for Diagnosis of Suspected Pulmonary Embolism (**ARTEMIS**) Trial (van der Pol et al, NEJM 2019; 380(12):1139-1149).

MCQ 2 2026

28F G2P1 currently 34 weeks gestation presents to the ER with acute pleuritic chest pain and dyspnea after a 9-hour transatlantic flight. On exam, HR 130bpm, BP 110/70, O2 sat 95% on RA. She appears to be mildly dyspneic, JVP is elevated, normal heart sounds, no crackles or wheeze, and no leg asymmetry. Routine blood work shows normal CBC, electrolytes, creatinine, Troponin and BNP. ECG shows sinus tachycardia. CXR is nil acute.

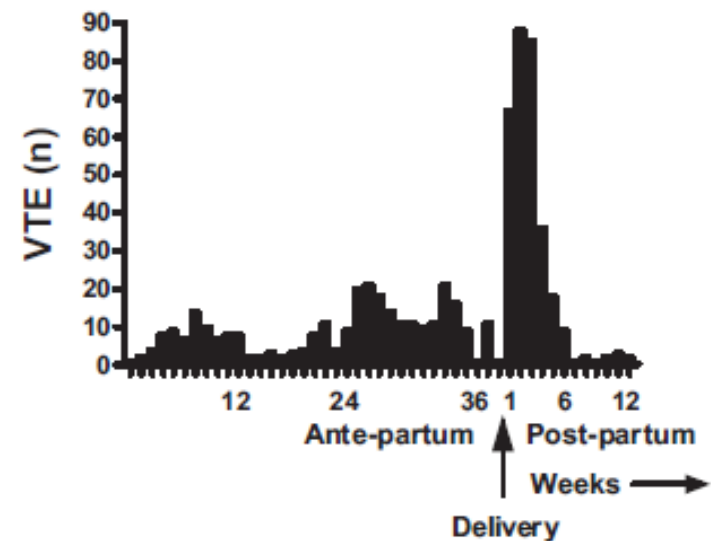
Which of the following is the least appropriate next step?

- a) Order a V/Q scan and if negative PE is ruled out
- b) Order a CTPA and if negative PE is ruled out
- c) Order a d-dimer and if <500ng/mL PE is ruled out
- d) Start therapeutic LMWH given high index of suspicion of PE
- e) Consult OB to assess fetus with ultrasound and heart rate monitoring

VTE

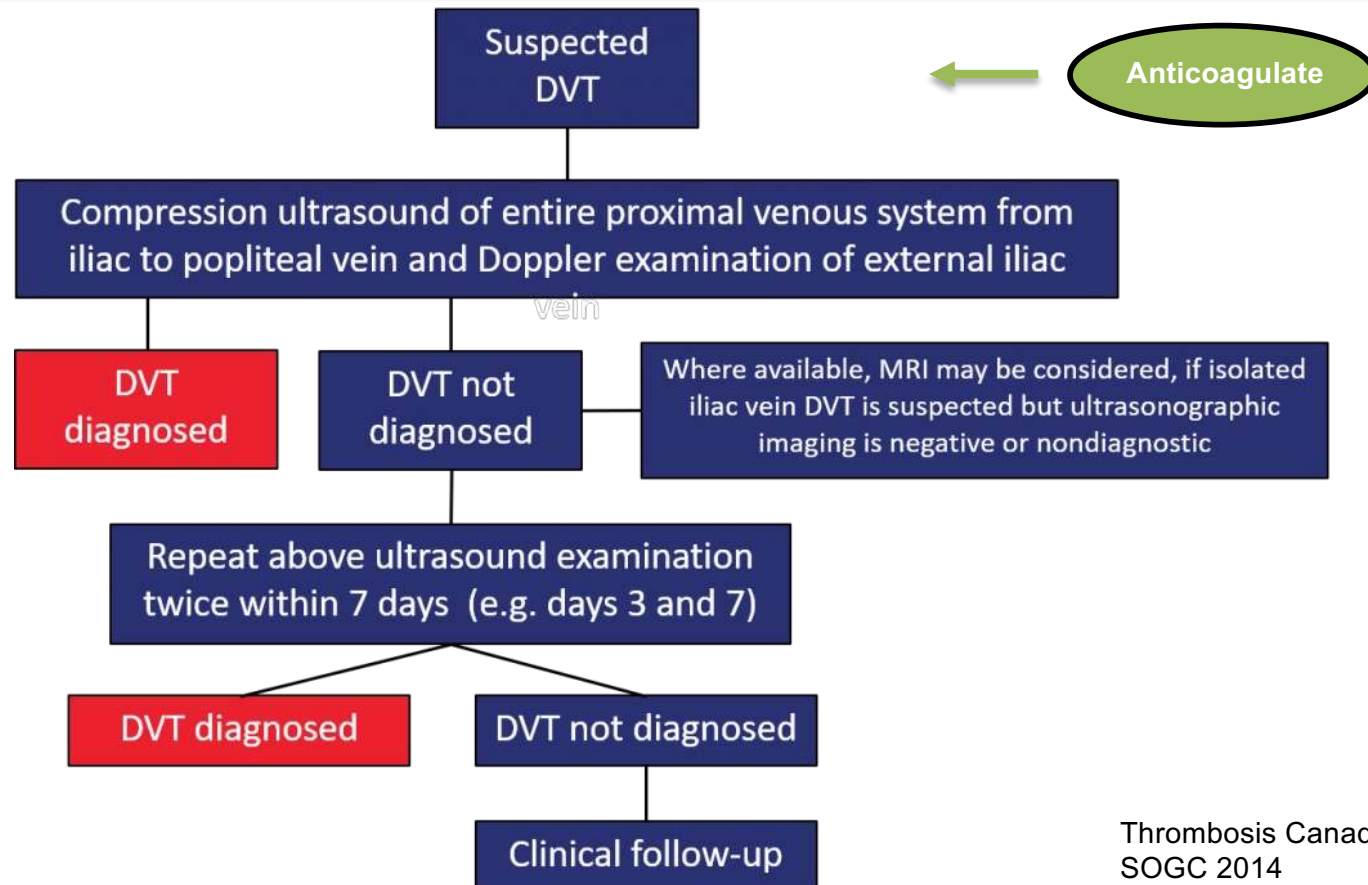
- Pregnancy increases risk of VTE
 - 5-10 fold increased risk during pregnancy
 - 5-35 fold in the **first 6 weeks post-partum**
- DVT
 - 1/1000 pregnancies
 - 80% occur in **left leg**
 - 60% isolated **iliac and femoral vein** (rather than popliteal)
 - Back/flank/buttock/thigh pain (rather than calf pain)
- PE
 - 0.5 per 1000 pregnancies
 - Hard to diagnose clinically – dyspnea and tachycardia can be normal in pregnancy

FIGURE 2
Distribution of VTE in pregnancy and puerperium



Anne AJOG 2008

Diagnosis of DVT



Thrombosis Canada 2021
SOGC 2014

BONUS
Read on own

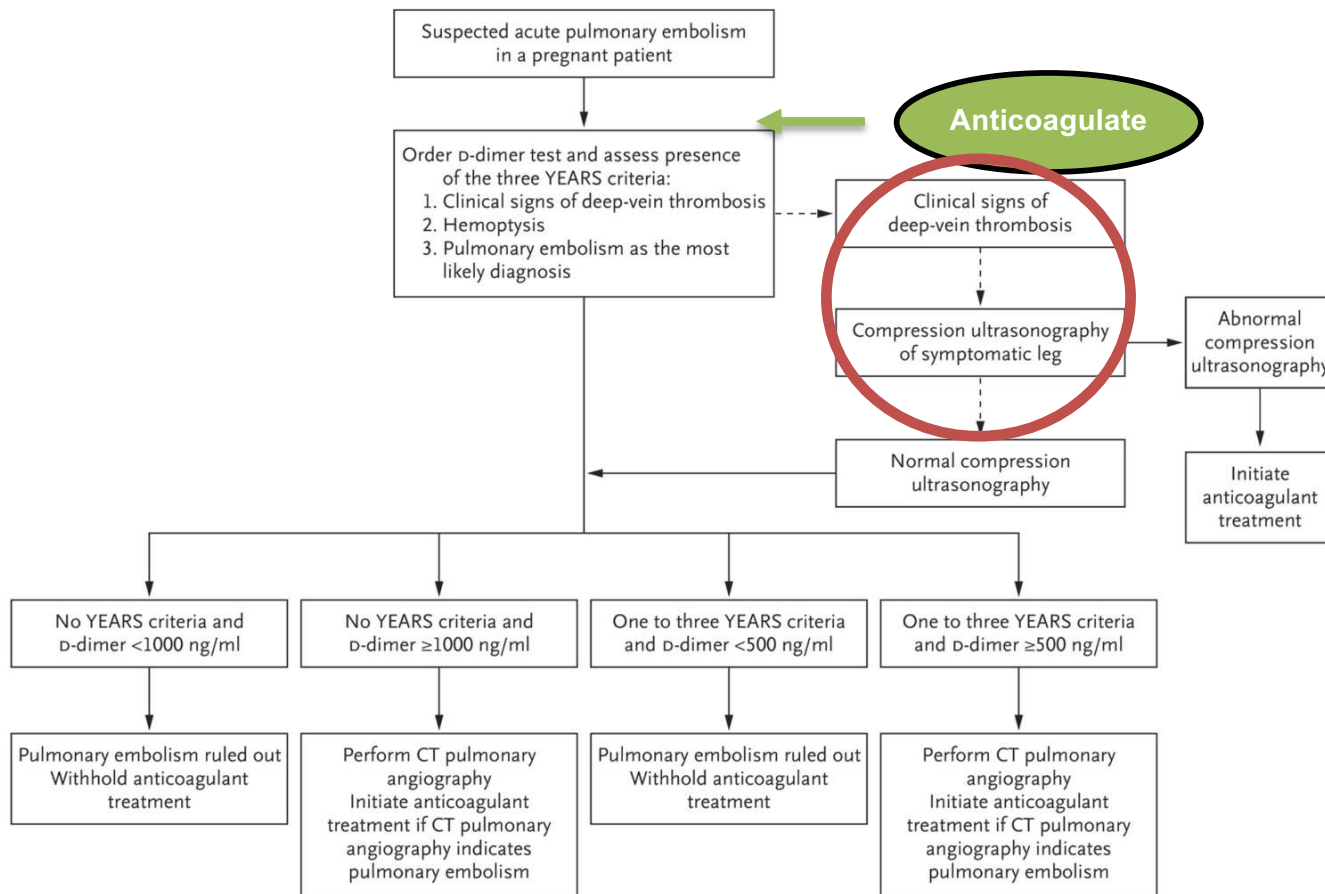
ARTEMIS Study (NEJM 2019 – PE Workup)

Prospective cohort, N=510, pregnant women 18+yrs presenting with chest pain or dyspnea, +/- tachycardia +/- hemoptysis presenting to ER or OB ward.

- Assess for 3 criteria of YEARS algorithm + D- Dimer.
 - 1) Signs or symptoms of DVT
 - 2) Hemoptysis yes/no
 - 3) PE felt to be most likely diagnosis by MRP (as assessed by Wells)
- 3 negative clinical criteria + d-dimer <1000 ng/mL: PE ruled out
- 1 or more clinical criteria + d-dimer <500 ng/mL: PE ruled out
- Those with signs of DVT underwent **leg ultrasound**, everyone else underwent **CTPA**
- Primary outcome: number of VTE at 3 months
- Secondary outcome: number of CTPA avoided
- OUTCOME:
 - only 1 DVT in follow-up, on day 90 of postpartum 90 d follow-up (false neg rate 0.21%; 95% CI 0.04 to 1.2). No PE
 - 39% of patients avoided CTPE (65% in first trimester), majority in first trimester, algorithm most useful in first trimester
 - in patients w/ high pretest probability (1+ YEARS criteria, + D-dimer >500), only 7.5% had PE Dx with CTPA

Meta-analysis of 4 trials including Artemis showed 99.5% sensitivity and **NPV of 100%** for d-dimer in pregnancy. It concluded that **d-dimer can safely rule out VTE** in pregnant women with a disease prevalence consistent with a low/intermediate or **unlikely pretest probability**. Bellesini et al J Thromb Haemost. 2021 Oct;19(10):2454-2467.

Diagnosis of PE



Thrombosis Canada 2021

It is essential that patients and physicians understand the **risks and consequences of not pursuing imaging**.

If you have a high pretest probability of PE, consider imaging.

Options for diagnostic imaging include V/Q scan and CTPA. **Some physicians prefer V/Q scanning.**

Postpartum patients should be investigated the **same as non-pregnant patients**. CT is safe in breastfeeding. Radioisotopes used in V/Q may contaminate breast milk for 24-48 hours.

Thrombosis Canada 2021

Radiation Counselling

- 1 rad = 10 milligray = 10mSv
- 10 rads required dose for teratogenicity or miscarriage
- Accept upper limit in pregnancy is **< 5 rad** (or 50 milligray)

Table 4

Fetal and Maternal Radiation Doses Associated with Diagnostic Tests for PE

Diagnostic Test	Fetal Dose (mGy)	Maternal Dose (Whole Body Effective Dose in mSv)
CXR	0.002	0.1
V/Q	0.32–0.74	1–2.5
CTPA	0.03–0.66	4–18
DSA	...	7–28

V/Q has << radiation for mother breast than CT (breast cancer), while slightly higher radiation to fetus (childhood cancer). Can do perfusion scan only for further reduced radiation.

Management of Acute VTE in Pregnancy

Duration: minimum **3 months** including **6 weeks post-partum**

Drug: **LMWH** (or UFH)

Considerations: Weight based dosing
Once or twice daily dosing
Xa level monitoring*
Risk of bleeding and HIT

(eg) enoxaparin 1mg/kg sc bid
dalteparin 200 u/kg sc daily
tinzeprin 175 u/kg sc daily

*[ASH 2018 recommends AGAINST routine anti-Xa level monitoring]

What about the other Anticoagulants?

- **Warfarin relatively contraindicated** in pregnancy (**OK during breastfeeding**)
 - First trimester: warfarin embryopathy (mid-facial and limb hypoplasia and stippled bone)
 - Second/third trimesters: neurologic anomalies, microcephaly, optic atrophy, neonatal hemorrhage
 - Nuances in mechanical valves discussed later in talk
- **NOACs not recommended** in pregnant or breastfeeding women (cross placenta)
- HIT: Danaparoid first line, Fondaparinux second line, consult heme
 - Fondaparinux has been shown to cross the placenta, experience limited esp in Trimester 1
- tPA
 - If life threatening and hemodynamically unstable from PE, or limb threatening from DVT
 - Does not cross placenta (large molecule)
- IVC filter
 - Same indications as outside of pregnancy (eg. concurrent bleeding, peri-operative)
 - Consider use if VTE is newly diagnosed closed to delivery (ie <2-4 weeks), prompt removal post- partum, hematology consultation

VTE Prophylaxis (SOGC, ASH): Patient has history of a Prior VTE

Clinical Situation	Antepartum	Postpartum 6 weeks
Prior VTE - Unprovoked - Estrogen related (pregnancy, OCP)	✓	✓
Prior provoked VTE (non-estrogen, reversible or temporary risk factor)	X	✓

****ASH 2023 Thrombophilia guidelines have a weak, conditional recommendation to test patients with hx pregnancy / OCP related VTE and if thrombophilia present “treat indefinitely with anticoagulants.”**

***This is NOT standard practice in Canada. To be discussed in your hematology lecture.**

Thrombosis Canada 2020

VTE Prophylaxis (ASH):

No Prior VTE *but clarify family history of VTE*

Clinical Situation	Antepartum	Postpartum 6 weeks
Homozygous FVL	✓	✓
Combined thrombophilia	✓	✓
ATIII deficiency (if no fam hx – why were they tested in the first place?) <i>SOGC doesn't stipulate family history, just says to prophylaxis during/after with ATIII deficiency)</i>	✓ if + fam history ✗ if no fam history	✓ if + fam history ✗ if no fam history
Homozygous PT20210A *no evidence, this was consensus of group	✓ if + fam history* ✗ if no fam history	✓
Protein C or S deficiency	✗	✓ 'suggests' if + fam history ✗ if no family history
Heterozygous FVL, heterozygous PT mutation	✗	✗ even if + fam history
APLA and history of recurrent pregnancy loss <i>From SOGC – not covered in Thrombosis Canada</i>	✓ +low dose ASA	✓

VTE Prophylaxis (ASH):

No Prior VTE *but clarify family history of VTE*

Clinical Situation	Antepartum	Postpartum 6 weeks
Homozygous FVL	✓	✓
Combined thrombophilias	✓	✓
ATIII deficiency (if no fam hx – why were they tested in the first place?) <i>SOGC doesn't stipulate family history, just says to prophylaxis during/after with ATIII deficiency)</i>	✓ if + fam history ✗ if no fam history	✓ if + fam history ✗ if no fam history
Homozygous PT20210A *no evidence, this was consensus of group	✓ if + fam history* ✗ if no fam history	✓
Protein C or S deficiency	✗	✓ 'suggests' if + fam history ✗ if no family history
Heterozygous FVL, heterozygous PT mutation	✗	✗ even if + fam history
APLA and OB-APS or hx thrombosis <i>From SOGC 2014 and ACR 2020 Reproductive</i>	✓ +low dose ASA	✓ (+ low dose ASA, consider 6-12 weeks PP (ACR2020))

More on Thromboprophylaxis

- **Other risk factors for VTE** (Thrombosis Canada 2020)
 - C-section with 1 major or 2 minor risk factors:
 - Major: preeclampsia with IUGR, prior VTE, high risk thrombophilia, antepartum bed rest >7 days...and more
 - Minor: BMI>30, multiples, smoking, IUGR, lower risk thrombophilia, preeclampsia, emergency c/s

[beyond scope – OB will discuss / team based decision]
- **Prophylactic Dosage**
 - Enoxaparin 40mg daily, Dalteparin 5000 U daily, Tinzaparin 4500U daily
 - For **patients >100kg**: Enoxaparin 40mg BID or 80mg daily, Dalteparin 5000 BID or 10,000mg daily, Tinzaparin 10,000U daily

Delivery

- Therapeutic anticoagulation
 - Planned delivery (induced vaginal delivery or C-section – *mode* determined by OB)
 - Withhold *therapeutic LMWH* for **24 hours** pre neuraxial analgesia
 - **Admit for UFH if VTE diagnosed <4 weeks of delivery**, hold 6 hours pre neuraxial analgesia / parturition
- Prophylactic anticoagulation
 - Planned or spontaneous labour (ASH 2018)
 - Withhold *prophylactic dose LMWH* anticoagulation for **12 hours** pre neuraxial analgesia
- **Withhold anticoagulation if in spontaneous labour**
- Post partum
 - Restart prophylactic anticoagulation >4 hours post removal of neuraxial anesthesia
 - Restart anticoagulation 4-6 hours post vaginal delivery, 6-8 hours post c-section, if hemostasis achieved

SOGC 2014, ASH 2018, ASRA 2010

Breast Feeding



- Medications that are safe if breastfeeding

- LMWH
- UFH
- Danaparoid
- **+ Warfarin**

- If not breast feeding

- All of above **+ NOACs**

Every prescription for anticoagulant must also be accompanied by a discussion of CONTRACEPTION

- Implicit is a need to avoid estrogen based contraceptives (VTE Risk, effects of milk supply)
- Progestin only pill (norethindrone) or Progestin IUD (Ex Mirena, Kyleena) or progestin implant (Ex Nexplanon) or surgical contraception (tubal ligation)



MCQ 2 2025

28F G2P1 currently 35 weeks gestation presents to the ER with acute pleuritic chest pain and dyspnea after a 9-hour transatlantic flight. On exam, HR 130bpm, BP 110/70, O2 sat 95% on RA. She appears to be mildly dyspneic, JVP is elevated, normal heart sounds, no crackles or wheeze, and no leg asymmetry. Labs: CBC normal, electrolytes, creatinine, Troponin are normal.

Scenario with high index of suspicion of pulmonary embolism

Which of the following is the least appropriate?

- a) Order a V/Q scan and if negative start empiric anticoagulation
 - b) Order a CTPA and if negative P start empiric anticoagulation
 - c) **Order a d-dimer and if <500ng/ml start empiric anticoagulation**
 - d) Start therapeutic LMWH given weight and renal function
 - e) Consult OB to assess fetus with ultrasound
- a. You can undergo V/Q testing if accessible and with normal baseline CXR to assess for PE in pregnancy
 - b. You can undergo CTPA for diagnosis of PE in pregnancy, counsel patients on higher maternal breast radiation and lower fetal radiation compared to V/Q scan
 - c. **Correct answer – d-dimer should only be used in low or intermediate risk probability patients, similar to its use in the general population. Imaging with CTPA or V/Q scan should be first line in this case where PE is highly likely.**
 - d. Start empiric anticoagulation where suspicion of PE is high while concurrently undergoing investigations
 - e. Don't forget to involve OB to monitor fetal status in a sick patient

Thrombocytopenia

- Top 3 differential diagnosis
 - 1) Gestational thrombocytopenia
 - 2) ITP
 - 3) Pre-eclampsia with HELLP
- Platelet targets
 - ITP: **Vaginal delivery >30, C-section >50, neuraxial anesthesia >70-80***
 - HELLP/Preeclampsia: <20 transfuse all; C section <50 transfuse; >50 – transfuse if actively hemorrhaging or rapidly falling counts with other coagulopathy
- Basic investigations
 - CBC, reticulocytes, blood smear, liver function tests, HIV/HBV/HCV, urinalysis/urine protein to creatinine ratio
 - Consider: TSH, ANA, APLA, H. Pylori, INR/PTT, DAT

****ITP platelet target safe for neuraxial anesthesia is center and anesthesiologist specific.*** Generally, avoid epidural when platelets are <50 given 9-11% risk of epidural hematoma.

ASH 2013

Thrombocytopenia

- Gestational thrombocytopenia (70-80%)
 - Decreasing platelet count in pregnancy worse in T3 or newly discovered thrombocytopenia
 - Typically platelets >70 (**generally >100**)
 - Diagnosis of exclusion
 - No treatment required, typically resolve <6 weeks PP
- ITP (1-4%)
 - Pre-existing thrombocytopenia, plt <100
 - Treat if platelets <30, clinically bleeding, or **<50 near delivery**
 - **Prednisone or IVIG**, avoid Dexamethasone as crosses placenta
 - Consult pediatrics to rule out neonatal thrombocytopenia (1-10%)
 - OB to avoid instrumentation ex vacuum extraction if maternal ITP due to this risk

More Details on ITP
in your Hematology
Slide Deck!

ASH 2013

Anemia in Pregnancy: Hb <110g/L

- Most common cause is iron deficiency
 - It takes 1000mg of iron to support fetal growth, placenta and increased maternal blood volume
 - Pregnant patients should have CBC and ferritin in the first trimester blood work
 - if Hb>110g/L and ferritin <50ug/L, start oral iron, repeat in second trimester and if Hb remains above 110g/L continue oral iron
 - If Hb<110g/L and ferritin <50ug/L, start oral iron and consider IV iron, repeat in second trimester and if persistent consider IV iron
 - WHO suggests starting oral iron in ALL pregnant patients
 - Check b12 if malabsorption syndrome, PPI use, severe Tri1 nausea/ vomiting / weight loss, metformin, normocytic anemia (should be microcytic if pure IDA) etc
 - If MCV <80 or MCH <27, Consider hemoglobinopathy/ sickle cell disease if patient of appropriate ancestry = Hb electrophoresis

Anemia in Pregnancy: Hb <110g/L

- Iron deficiency anemia in pregnancy associated with:
 - Increased risk of Postpartum hemorrhage (PPH)
 - Maternal anxiety and depression
 - Fatigue, shortness of breath
- **IV iron** (iron sucrose) safe and indicated antepartum / postpartum for severe symptomatic anemia, or anticipated high risk delivery for PPH
 - Eg concomitant thrombocytopenia or bleeding disorder, peripartum anticoagulants, obesity, operative delivery

Reduce risk of postpartum hemorrhage by:

- Correcting iron deficiency anemia antepartum
- Intrapartum **TRANEXAMIC ACID** if risk of PPH = ↓ bleeding and does not increase VTE risk
- Prophylactic TXA post vaginal delivery does NOT ↓ risk of PPH (N Engl J Med. 2018 Aug 23;379(8):731-742) and post C-section did not reduce rates of blood transfusions or maternal mortality (N Engl J Med. 2023;388(15):1365.)

Outline

- Hypertension & Pre-eclampsia
- VTE and ITP in pregnancy
- **Diabetes in pregnancy**
- High Yield GI disorders of pregnancy
- High Yield Cardio in pregnancy
- High Yield Resp in pregnancy
- BONUS SLIDES
 - High Yield Infections in pregnancy
 - High Yield Heme disorders of pregnancy
 - Drugs in Pregnancy
 - Approach to the Obstetric Patient in a Scenario
 - Pre-pregnancy counselling for women with chronic conditions
 - Normal Labs in Pregnancy
- COVERED ELSEWHERE:
 - Thyroid → ENDO
 - Asthma Treatment → RESP
 - HIV and others in pregnancy → ID

Diabetes in Pregnancy

- Key Resources:
 - **Diabetes Canada Guideline.** Feig DS, et al. Can J Diabetes. 2018 Apr;42 Suppl 1:S255-S282.
 - **CONCEPTT trial:** Feig DS, et al. Lancet. 2017 Nov 25;390(10110):2347-2359.
 - **MiTY trial:** Feig DS, et al. Lancet Diabetes and Endocrinology. 2020 Oct 1; 8(10).

MCQ 3 2025

32F G1P0 currently 24 weeks of gestation. She is previously healthy and has an uncomplicated pregnancy.

She underwent a OGCT with a result of 12.1.

Which of the following statement is false?

- a) Gestational diabetes is diagnosed
- b) Start monitoring daily fasting and 2-hour postprandial blood sugars
- c) NPH insulin can be started if she has elevated fasting blood sugars
- d) Metformin and Glyburide may be used as alternatives to insulin
- e) Start Aspirin 162mg daily now and continue until 36 weeks

Effects of Diabetes on Pregnancy

Fetal

- **Teratogen:** hyperglycemia can cause congenital cardiac malformations, spinal cord defects, cleft lip/palate [if severe trimester 1 hyperglycemia, not GDM]
 - Enhanced cardiac screen / fetal echo if poor glycemic control with pregestational diabetes
- Miscarriage, macrosomia, shoulder dystocia, pre-term labour, stillbirth
- Neonatal: hypoglycemia, hypocalcemia, hyperbilirubinemia

Maternal

- Exacerbating microvascular complications, DKA, bladder infection, operative delivery (ie C-section), pelvic injury during vaginal delivery, hypertensive disorders including pre-eclampsia

Preconception Counselling for type 1/2 diabetes

- Planned pregnancy with **HbA1c<7%, ideally <6.5%**
- Folate 1 mg for 3 months pre- and 12 weeks post-conception
- Screen for retinopathy, nephropathy (Cr, ACR), blood pressure, HbA1c, +/- ECG or ECHO
- **Discontinue** contraindicated medications
 - **ACEi/ARB** (if diabetic nephropathy, can continue up until pregnancy)
 - **Statin** (if trying to conceive – **discuss with cardio if for secondary prevention**)
 - Non-insulin oral hypoglycemic agents, except for Metformin and Glyburide
 - There is no human data on the use of DPP-4 inhibitors, GLP-1 receptor agonists or SGLT2 inhibitors. The use of these non-insulin antihyperglycemic agents is not recommended in pregnancy thus should **stop** prior to pregnancy [guidelines.diabetes.ca/cpg/chapter36]
- For both type 1 and 2 diabetes, start **ASA 162 mg daily at 12-16 weeks until 36 weeks** to decrease the risk of pre-eclampsia

Type 1 Diabetes

- MDI or insulin pump
- Consider continuous glucose monitoring (CGM)
- **CONCEPTT trial** (Feig et al. Lancet 2017): CGM reduces HbA1c, more time in glycemic target, reduced fetal LGA, fetal hypoglycemia and NICU admissions. No difference in rates of maternal. Hypoglycemia. Most common side effect were skin reactions.

Type 2 Diabetes

- May continue **metformin or glyburide**
- If on other oral antihyperglycemic agents, switch to **insulin** prior to conception
- **SGLT-2i and GLP-1 agonists are not recommended** for use in pregnancy or breastfeeding
- **MiTy Trial** (Feig et al. Lancet 2020) type 2 diabetes of Metformin vs placebo: No difference in composite neonatal morbidity and mortality. Mothers had better glycemic control, less weight gain, and less requirement for insulin. Lower infant weight noted in secondary outcome. Long term effects on infants/children weren't studied, paucity of data in literature.

Gestational Diabetes

Screening before 24 weeks is not yet recommended in Canada. We acknowledge ACOG suggests earlier screening if significant risk factors (e.g. Morbid obesity, prior GDM, A1C $\geq 5.7\%$...) and there may be benefit to this (TOBOGM trial NEJM 2023). This is emerging data not expected to be testable in our exam!

Screening at **24-28 weeks**

50g 1hr **OGCT**: ≥ 11.1 GDM diagnosed

7.8-11.0 Borderline – move to 2hr 75g OGTT

Less than 7.8 Normal

75g 2hr **OGTT**:

Cut-off if used as 2nd step (e.g. after borderline 50g) FPG ≥ 5.3 , 1hr ≥ 10.6 , 2hr ≥ 9.0

Cut-off if used as 1 step screen (not recommended / alternate) FPG ≥ 5.1 , 1 hr ≥ 10.0 , 2 hr ≥ 8.5
if 1 value is met or exceeded, diagnosis of gestational diabetes is made

If high risk of T2DM, screen earlier with HbA1C or FBG. If early screening is negative, screen again with OGCT or OGTT at 24-28 weeks.

Gestational Diabetes

BG Targets same as pre-existing diabetes

Treatment:

- **Lifestyle** Interventions- Diet and Exercise
- First Line: **Insulin NPH** (to target FBG) +/- rapid acting insulin (to target post-prandial BG)
- Metformin may be used as an alternate to insulin
 - 40% on metformin will still require insulin
 - Inform the patient that metformin crosses the placenta (as does insulin. Not a teratogen)
- Glyburide may be used in combination with metformin if not reaching targets and refusing insulin (not Gliclazide)

Blood Glucose Targets

BS monitoring

- 4 times daily: fasting, one *or* two hours post meals

BG Targets

- **Fasting <5.3**, 1h post-prandial <7.8, **2h post-prandial <6.7**
- HbA1C <6.5% in pregnancy
- Intrapartum BG maintained between 4.0-7.0 (to minimize neonatal hypoglycemia)

Postprandial = 1 or 2 hrs after first bite of the meal

Postpartum

Pre-existing diabetes

- **Decrease insulin** to below pre-pregnancy dose (more insulin-sensitive)
- Metformin and glyburide may be used in breastfeeding
 - Caution in neonatal period if concerns re prematurity, renal impairment
- **Resume ACEi** for diabetic nephropathy (e.g. Enalapril during breast feeding)
- Screen for post-partum thyroiditis in T1DM approx. 3 mos postpartum

Gestation diabetes

- **STOP insulin/metformin/glyburide** postpartum
- **75g OGTT at 6 weeks** – 6 months postpartum to rule out T2DM
- Encourage breastfeeding immediately to prevent neonatal hypoglycemia
- 50% of women will have GDM again in subsequent pregnancies; 20% of women will develop T2DM in 10 years

MCQ 3 2025

32F G1P0 currently 24 weeks of gestation. She is previously healthy and has an uncomplicated pregnancy.

She underwent a OGCT with a result of 12.1.

Which of the following statements is correct?

- a) Gestational diabetes is diagnosed by a fasting glucose of ≥ 126 mg/dL
- b) Start monitoring daily fasting glucose
- c) NPH insulin can be started if fasting glucose is ≥ 126 mg/dL
- d) Metformin and Glyburide are safe oral hypoglycemics in pregnancy
- e) **Start Aspirin 162mg daily**

Gestational diabetes

- a. 50g 1hr OGCT of ≥ 11.1 is diagnostic of GDM
- b. Glucose monitoring in pregnancy include fasting and either 1-hour or 2-hour postprandial glucose
- c. NPH insulin is first line for gestational diabetes +/- rapid acting insulin
- d. Metformin and Glyburide are safe oral hypoglycemics in pregnancy
- e. **Correct answer: aspirin for pre-eclampsia prevention is only indicated for those with pre-existing type 1 or 2 diabetes, not for gestational diabetes**

Outline

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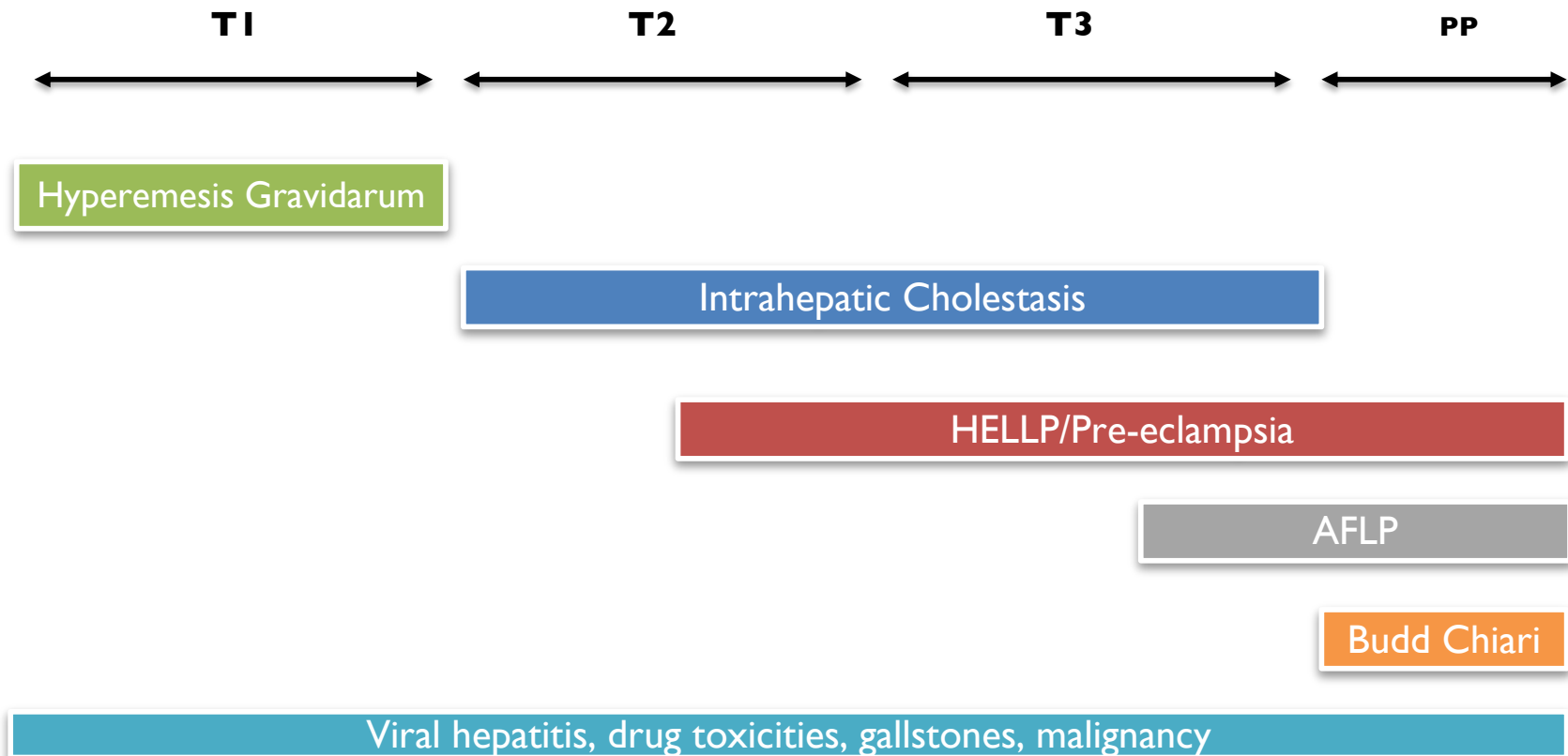
MCQ 4 2025

32-year-old female G1P0 at 32 weeks of gestation presents with pruritus of hands and feet worse at night for two weeks. She denies any history of renal, liver or autoimmune diseases. No rashes. Blood work shows non fasting bile acid of 65, AST 230, ALT 350, Bilirubin 10. Abdominal ultrasound shows normal liver, gallbladder, and patent biliary tract. Hepatitis BsAg, core antibody and Hepatitis C antibody are negative.

Which of the following is false?

- a) She has intrahepatic cholestasis of pregnancy
- b) Repeat bile acid and liver enzymes every 2 weeks
- c) Start Ursodeoxycholic acid 500mg po BID
- d) Reassure her that this condition causes itch without any fetal complications
- e) Counsel her on the high risk of recurrence in future pregnancies

Liver Diseases in Pregnancy



Liver Disease

	Hyperemesis Gravidarum	IHCP Intrahepatic Cholestasis	HELLP +/- pre-eclampsia	AFLP Acute fatty liver of pregnancy
Gest'n	T1	T2-3	> 20 weeks	T3 (late)
Symp-toms	Vomiting >3x/day, 5% weight loss Dehydration	Pruritus, <u>without rash</u> , starts in palms and soles, worse at night Risks: other liver dz (ex fatty liver, hep C)	RUQ pain, nausea/vomiting Symptoms of pre-eclampsia	Vomiting, abdominal pain, encephalopathy, jaundice, ascites, <u>polydipsia and polyuria</u>
AST,ALT	100's	Several 100's	1000's	< 500
Bili-rubin	Normal	Normal or mildly elevated	↑ unconjugated bili 2o hemolysis	↑↑↑ conjugated bili
Other tests	Ketosis, electrolyte imbalance	NON-Fasting bile acid* >19 umol/L (repeat in 2-4 weeks if initially normal and typical sx's) Bile acid >40 = ↑ meconium-stained amniotic fluid, preterm birth, neonatal resp distress, NICU admits Bile acid >100 = ↑ stillbirth ↑Risk of GDM, HTN, Preeclampsia w IHCP!	Platelets <100 (early finding) Microangiopathic hemolytic anemia (late finding) Abdominal ultrasound (r/o hepatic hemorrhage, infarct and rupture)	Leukocytosis, hypoglycemia, elevated ammonia, renal failure, ascites on ultrasound, microvesicular steatosis liver biopsy (Swansea Criteria)

*SOGC 2024 Practice Guideline No. 452: Diagnosis and Management of Intrahepatic Cholestasis of Pregnancy

Liver Disease : Treatment

	Hyperemesis Gravidarum	IHCP Intrahepatic Cholestasis	HELLP +/- pre-eclampsia	AFLP Acute fatty liver of pregnancy
GA	T1	T2-3	> 20 weeks	T3 (late)
Treat ment	Anti-emetics, fluids, electrolyte replacement	→ Monitor for complications (Delivery by 36 weeks if bile acids >100) → Ursodeoxycholic acid (10-15mg/kg/d in 2-3 divided doses) safe, may help itch and transaminitis, but recent RCT (PITCHES Lancet 2019) suggests not effective to reduce adverse neonatal outcomes (ie stillbirth) → Itch adjuncts: Hydroxyzine, Rifampicin	Delivery BP control Eclampsia prevention	Delivery Liver transplant if hepatic failure Screen newborn for LCHAD

Nausea/Vomiting

- **Morning Sickness** (50-80%) (SOGC Dec 2016 Guideline)
 - NON-PHARMACOLOGIC:
 - Stop iron containing prenatal, take folic acid only
 - Diet – small frequent meals, fluids ‘cold/clear/carbonated’
 - Ginger (I-A) 250mg po QID, Acupressure (I-B), Mindfulness Retraining
 - PHARMACOLOGIC:
 - First Line Pyridoxine (Vitamin B6) or Pyridoxine-doxylamine (**Diclectin**) (I-B)
 - 2nd Line H1-blockers (ex **Gravol**) (I-A), then: chlorpromazine (I-A), **Metoclopramide** (II-2B), then ondansetron (II-1C, check ECG re QTc)
 - IV hydration supplemented w/ multivitamins (ACOG guideline)
- **Hyperemesis Gravidarum** (HG) (0.3-2%)
 - Diagnosis (Fairweather criteria, Windsor definition): vomiting >3x daily, dehydration, electrolyte imbalance, ketosis/ketonuria, >5% weight loss, transaminitis
 - Treatment: Metoclopramide + Gravol + PPI acutely, replace electrolytes and fluids
- **GERD** (**common cause of nausea in pregnancy!) [SOGC 2022]
 - Very common; treatment: Calcium carbonate (constipating) → H2 blocker → PPIs
- **Marijuana use** – counsel pregnant/lactating/pre-conception women to stop using all cannabis products

BONUS

Read on own

Hepatitis in Pregnancy

- Hepatitis B
 - All pregnant women tested for hepatitis B surface antigen
 - Risk of mother to child transmission, especially with invasive procedures (e.g. amniocentesis)
 - Antiviral therapy started if maternal HBV DNA >200,000 IU/mL *[see GI lecture for more info!]*
 - Tenofovir is preferred treatment, started between 28-32 weeks
 - Breastfeeding is not contraindicated, even those on antiviral therapy
 - Infant should receive Hepatitis B vaccine and IVIG (*CONSULT PEDIATRICIAN, REFER BABY TO LOCAL HEALTH UNIT FOR VACCINE SERIES – needs more shots at 2, 6 mos*)
- Hepatitis C (SOGC 2025 Consensus Statement)
 - **Hepatitis C antibody screening should be performed in the first trimester of every pregnancy**
 - Hepatitis C viral load should be done in those with a history of hepatitis C infection (treated or untreated)
 - Mother to child vertical transmission can occur intrapartum, peripartum, or postpartum
 - No indication for C-section or other changes to routine intrapartum care except avoidance of procedures mixing fetal and maternal blood, such as fetal scalp electrodes and fetal blood sampling
 - Treatment for hepatitis C is not recommended during pregnancy or breastfeeding (insufficient data on antivirals to recommend use in pregnancy). Patients should be referred for treatment following delivery prior to discharge from hospital with baseline liver function tests and assessment of fibrosis
 - Breastfeeding is not contraindicated but should be avoided in setting of cracked or bleeding nipples
 - Neonates should be tested for HCV at 2 months of age
 - Preconception antiviral therapy prioritized for HCV infected women of childbearing age

MCQ 4 2025

32-year-old female G1P0 at 32 weeks of gestation presents with pruritus of hands and feet worse at night for two weeks. She denies any history of renal, liver or autoimmune diseases. No rashes. Blood work shows non fasting bile acid of 65, AST 230, ALT 350, Bilirubin 10. Abdominal ultrasound shows normal liver, gallbladder, and patent biliary tract. Hepatitis B and C are negative.

Which of the following is false?

- a) She has intrahepatic cholestasis
 - b) Repeat bile acid and liver enzymes
 - c) Start Ursodeoxycholic acid 500mg daily
 - d) Reassure her that this condition is benign**
 - e) Counsel her on the high risk of stillbirth
- Intrahepatic cholestasis of pregnancy
- a. Diagnosed by pruritus, primarily of the hands and feet starting in late second and third trimester, and non-fasting bile acids of $>19\mu\text{mol/L}$
 - b. Regular lab reassessment to monitor for peak bile acid level
 - c. Ursodeoxycholic acid is first line treatment to reduce maternal pruritus. It will also improve transaminitis and decrease preterm birth and meconium-stained amniotic fluid, but does not prevent stillbirth.
 - d. Correct answer: cholestasis of pregnancy is a fetal condition with many fetal risks with rising bile acids. The most devastating is a $>3\%$ chance of stillbirth when bile acids exceeds $100\mu\text{mol/L}$ anytime during the pregnancy**
 - e. There is a 70-90% risk of recurrence of intrahepatic cholestasis in future pregnancies

Outline

- Hypertension & Pre-eclampsia
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- **High Yield Cardio in pregnancy** →
- High Yield Resp in pregnancy
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 - Thyroid → ENDO
 - Asthma Treatment → RESP
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CCS 2021 Update: Clinical Practice
Update on CV Management of the
Pregnant Patient

ACC/AHA 2023 Chronic CAD
Guideline with pregnancy
recommendations

MCQ 5 2025

39F G2P1 currently 30 weeks of gestation presents with new onset palpitations in the last month. She endorses 30 seconds to a minute of palpitations occurring once to twice daily. No associated chest pain or syncope. She is previously healthy without any history of arrhythmias or family history of heart disease.

48-hour Holter showed an average heart rate of 110bpm. There are 2% PACs, 10% PVCs, and occasional bigeminy. No pauses or other arrhythmias. Symptom diary showed an episode of palpitations correlating with a run of PVCs.

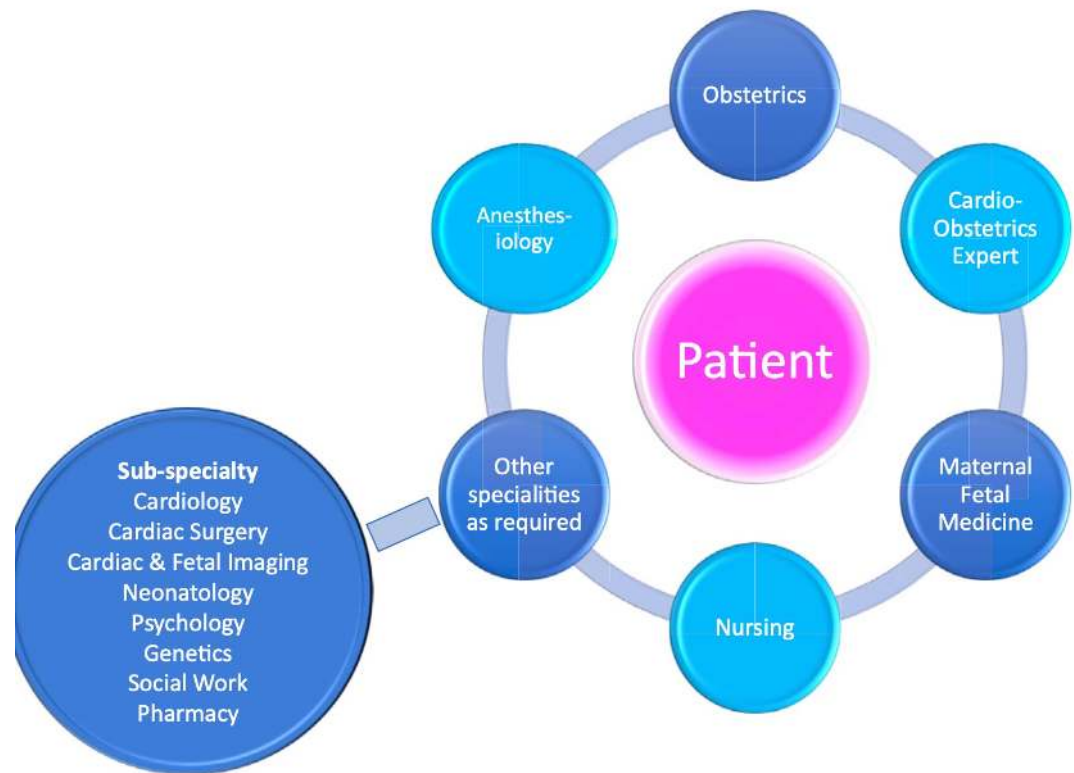
Echocardiogram shows normal biventricular function. LVEF of 60%. No hemodynamically significant valvulopathy.

Which of the following is the next best step?

- a) Start Labetalol 200mg po BID
- b) Start Metoprolol 25mg po TID
- c) Start Atenolol 25mg po daily
- d) Forgo testing for CBC, ferritin, electrolytes, calcium profile, and TSH as you already have a diagnosis
- e) Anticipate up-titration of betablocker postpartum

Cardiovascular diseases in pregnancy

- Congenital heart disease + mechanical valves requires management in specialized centres
- As our demographics change, cardiovascular diseases need to be included in standard differentials of pregnancy related complaints
- As general internists, remember to counsel women of childbearing age appropriately + **discuss contraception** + refer for subspecialty mgmt if planning preg



CCS 2021 Clinical practice update on cardiovascular management of the pregnant patient. Similar to ACC/AHA Chronic CAD Guideline 2023 which has nice section on Pregnancy

ACC/AHA 2023 Chronic CAD Guideline

– Pregnancy Notes

- Be aware that a history of heart disease puts women at very high risk of cardiac events in pregnancy
- Women with Chronic CAD have high rate of adverse maternal, fetal, neonatal events in pregnancy

Statins ... discussion if secondary prevention
 FDA recently (2021) removed black box warning re statins in pregnancy (still C/I in lactation).
 Limited data re: safety but maternal risks must be considered if discontinuing.

TABLE 20 CARPREG II Risk Prediction Model

CARPREG II Predictors	Points
Previous cardiac event or arrhythmia	3
Baseline NYHA functional class III to IV or cyanosis	3
Mechanical valve	3
Ventricular dysfunction	2
High-risk left-sided valve disease and LVOT obstruction	2
Pulmonary hypertension	2
CAD	2
High-risk aortopathy	2
No previous cardiac intervention	1
Late pregnancy assessment	1
CARPREG II Score	Predicted Risk, %
0 to 1	5
2	10
3	15
4	22
>4	41

Peripartum Cardiomyopathy

- Normal Pregnancy physiology
 - Cardiac output and plasma volume increases by 50%, peak in T2, decline by 2 weeks postpartum but can be up to 6 months post-partum
 - Common physiologic symptoms in pregnancy: dizziness, palpitations, dyspnea, orthopnea, peripheral edema
 - **Physiologic dyspnea of pregnancy** (up to 75%), due to progesterone causing hyperventilation, onset first or second trimester, mild and not progressive, not activity limiting
 - Hypoxia is abnormal!
 - Sinus tachycardia, prominent or mildly elevated JVP, displaced apex (due to elevated diaphragm), S3 common (up to 80% of women), occasional S4, soft systolic flow murmur, varicose veins and peripheral edema
- Definition:
 - Systolic dysfunction EF<45%
 - TIMING: **last month of pregnancy until 5 months post-partum**
 - In absence of an alternative etiology
- Differential of pulmonary edema in pregnancy
 - pre-eclampsia (non-cardiogenic, hypertensive), ischemic, arrhythmia, amniotic fluid embolism, sepsis...

Peripartum Cardiomyopathy

- Treatment:
 - Diuretics: **Lasix**, if presence of pulmonary edema, use with caution given risk of excessive volume contraction and reduced placental perfusion
 - Beta-blocker: continued or initiated during pregnancy, **Metoprolol** preferred over Labetalol as cardio-selective, use once out of acute heart failure, **avoid Atenolol**
 - Other agents: **Nitrates** (venous and arterial dilation), **Hydralazine** (if concurrent hypertension), **Digoxin** (refractory heart failure despite vasodilator and diuretics)
 - Avoid ACE/ARB in pregnancy, postpartum can use Enalapril, Captopril, or Quinapril in breast feeding
 - Avoid Spironolactone in pregnancy, ok in breast feeding
 - ARNI have not been studied in pregnancy, so use is not recommended
- Prognosis: 1/3 recover spontaneously postpartum
 - Consider Genetics referral. Should be seen in tertiary centre.
- Subsequent pregnancy counselling + contraception
 - If EF recovers >40% = 20% recurrence of peripartum cardiomyopathy
 - If EF remains <40% = 50% recurrence of peripartum cardiomyopathy and 20% associated mortality

CCS 2017

Syncope in Pregnancy

- 4-5% of pregnancies
- Pregnancy physiology exaggerates **orthostatic and vasovagal** responses
- Approach as you would any other syncope
 - Good history, physical, check ECG
 - ECHO / rule out structural heart disease
- If sick (in ED / Delivery Suite, hemodynamically unstable)
 - **Embolism** (PE, amniotic fluid)
 - Arrhythmia
 - Hemorrhage/Anemia
 - Think Abruption if abdo pain or vag bleeding
 - Exacerbation of congenital heart disease

Arrhythmias in Pregnancy

Arrhythmia	Changes in Pregnancy	Notes on Management
SVT	SVT can become more frequent/symptomatic in pregnancy, may require daily suppressive Rx	Adenosine safe CCB (verapamil / dilt less data but considered safe), Beta blockers safe, Cardiovert if need be
Atrial Fibrillation / Flutter	Rest HR higher, if symptomatic treat Anticoagulation issues in pregnancy – discuss risk/benefit and make delivery plan	No warfarin Tri 1 Digoxin, beta blockers safe DC cardiovert if unstable (w/ fetal HR monitor)/ ECHO mom for congen heart disease; ?TSH, ?OSA
VT / VF	Not common, thankfully Rx: Beta-block aggressively	Avoid Amiodarone = concerns re fetal hypothyroidism
AV blocks	May be asymptomatic in pregnancy but vagal tone of delivery = risk of higher block.	-Continuous cardiac monitoring in labour, Watch fluids in delivery – pulm edema risk -ECHO mom for congen heart disease (VSD)

Cardiac Drugs in Pregnancy Table adapted from CCS

Class of Medication	Drugs that are considered SAFE	Drugs with limited data, use with caution	Drugs that are considered CONTRAINDICATED in pregnancy	Drugs that are considered CONTRAINDICATED in breastfeeding
Arrhythmia **dig levels unreliable in preg	Adenosine, lidocaine, bisoprolol, metoprolol, nadolol, propranolol, digoxin**	Diltiazem, flecainide, procainamide, propafenone, sotalol, verapamil	Amiodarone, atenolol, ivabradine	Amiodarone, ivabradine
HTN / heart failure **amio: fetal thyrotoxicity, use only if all else fails	Labetalol, methyl-dopa, metoprolol, nifedipine, bumetanide, carvedilol, furosemide, dobutamine, dopamine, norepi	Amiodarone**, hydralazine, nitrates, nitroprusside, HCTZ, metolazone, milrinone, torsemide	ACE-I, ARB, MRA, ARNI, SGLT2i	ACEi other than captopril, enalapril, lisinopril ARB, ARNI, SLGT2i
Anticoagulants	ASA, LMWH, uFH	Clopidogrel, ticagrelor, warfarin, argatroban, bivalirudin, fondaparinux, TPA, TNK	DOACS (+Warfarin except for mech valves, see next slides!)	DOACS
Pulm HTN / Others		Epoprostenol, sildenafil, iloprost	Statins, bosentan	Statins, bosentan

Mechanical Heart Valves (AHA/ACC Valve 2020)

FYI – beyond GIM scope
other than to know to send
to high-risk OB and Cardio!

- Very **high risk of valve thrombosis** (5%) and **maternal mortality** (1%)
 - women who cannot maintain therapeutic anticoagulation throughout pregnancy including monitoring of anticoagulation should be counselled to avoid pregnancy (Class I recommendation)
- **Warfarin** is agent of choice to prevent thrombosis (preferred over LMWH), but crosses placenta and dose dependent complications:
 - First trimester: warfarin embryopathy 5-7%
 - Second/third trimesters: neurologic anomalies, microcephaly, and optic atrophy, neonatal hemorrhage
- Continue Warfarin in pre-conception period until pregnancy confirmed
 - If on Warfarin $\leq 5\text{mg per day}$ - **may continue throughout pregnancy (seek expert advice!)**
- All women on LMWH: bid dose-adjusted, targeting anti-Xa level 0.8-1.2 u/ml 4-6h post dose
- All women with prosthetic valves (+ high risk unrepaired severe valvular heart disease) should undergo pre-pregnancy **counseling by a cardiologist** with expertise in pregnancy. TTE should be done on all patients before pregnancy. All women need **a planned delivery at centre with expertise in managing anticoag and valvular heart disease**

BONUS
Read on own

Mechanical Heart Valves (AHA/ACC Valve 2020)

FYI only. Not likely testable, but it is “good to know” that low dose warfarin is considered safe in these women at very high risk of thrombotic events.

- Warfarin $\leq 5\text{mg}$:
 - Continue warfarin throughout all 3 trimesters (Class IIa) or
 - dose-adjusted LMWH trimester 1 followed by warfarin Trimester 2, 3 (Class IIb)
- Warfarin $> 5\text{mg}$:
 - dose-adjusted LMWH trimester 1 followed by warfarin Trimester 2, 3 (Class IIa) or
 - dose adjusted LMWH all three trimesters (Class IIb)
- Delivery:
 - All patients need a **planned delivery**
 - Avoid delivery on VKA, due to high risk of fetal intracerebral hemorrhage. VKA should switch to **dose-adjusted bid LMWH** or IV UFH at **least 1 week** before planned delivery (CLASS I)
 - Women on LMWH must be switched to **IV UFH at least 36h prior** to planned delivery (CLASS I)
 - Stop UFH at least 6h prior to planned vaginal delivery (CLASS I)
 - If labour begins or urgent delivery required in woman on VKA, cesarean section should be performed after reversal of anticoagulation (CLASS I)
 - Bridge back to Warfarin postpartum (safe during breastfeeding)

BONUS
Read on own

Valvular Heart Disease (VHD) in Pregnancy (AHA/ACC Valve 2020)

- Severe stenotic VHD = High risk of heart failure, arrhythmia, intrapartum instability.
- Pre-pregnancy evaluation (**TTE** + consult with **cardiologist expert** in managing VHD in pregnancy) recommended (Class I) for all women with known or suspected valvular disease considering pregnancy
 - **Symptomatic severe** STENOTIC lesions should have **surgery** pre-preg (Class I)
 - Asymptomatic severe STENOTIC lesions and asymptomatic severe MR suitable for repair should be considered for intervention pre-pregnancy (Class II)
 - REGURGITANT lesions are generally better tolerated in pregnancy

Respiratory Disorders in Pregnancy

- Community Acquired Pneumonia
 - Increased risk of preterm birth, pre-eclampsia, low birth weight
 - First-line antibiotics : **Macrolide or beta-lactam**, Oseltamivir and flu, pneumococcal vaccines safe
 - **FQs** in animal model associated w/ cartilage problems
 - Where appropriate (ex exacerbations in pts with CF!), you can use these however!! Counsel.
 - No increased SA or Fetal malformations in cohort of women exposed to FQ in Tri 1. Non signif increase in malformation in moxifloxacin group; cipro gati levo no difference.
 - » Antimicrob Agents Chemoth 2014 58:8(4392-4398).
 - **Tetracyclines** – fetal teeth staining, bone growth suppression
 - **Sulfa** - kernicterus
- Asthma – same treatment as non-pregnant, **steroids are safe (po and inhaled)**
- OSA in pregnancy – gets worse due to N physiologic changes of pregnancy; associated with adverse maternal outcomes. No guidelines on screening/treatment in pregnancy (consensus = treat mod-severe, consider screening if at-risk pregnancy (ex early <34wks gestational hypertension in obese woman))

MCQ 5 2025

39F G2P1 currently 30 weeks of gestation presents with new onset palpitations in the last month. She endorses 30 seconds to a minute of palpitations occurring once to twice daily. No associated chest pain or syncope. She is previously healthy without any history of arrhythmias or family history of heart disease.

48-hour Holter showed an average heart rate of 75 bpm with occasional premature ventricular contractions (PVCs) in pregnancy. No bigeminy. No pauses or other arrhythmias. No runs of PVCs.

Echocardiogram shows normal biventricular function and no significant valvulopathy.

Which of the following is the next best step?

- a) Start Labetalol 200mg po BID
 - b) Start Metoprolol 25mg po TID
 - c) Start Atenolol 25mg po daily
 - d) Forgo testing for CBC, ferritin, electrolytes
 - e) Anticipate up-titration of betablocker
- a. Labetalol is a non-cardioselective betablocker so should not be used for arrhythmias in pregnancy. It is a first line antihypertensive given its alpha receptor blockade and reduction in peripheral vascular resistance.
 - b. **Correct answer:** PVCs are a common cause of palpitations in pregnancy. Patients who are symptomatic or have a high PVC burden should be treated with cardioselective betablockers, such as metoprolol or bisoprolol.
 - c. Avoid atenolol given risk of small for gestation fetus and neonatal bradycardia and hypoglycemia.
 - d. Do secondary testing for metabolic causes of arrhythmias and correct the abnormalities.
 - e. Anticipate need for rapid down-titration of betablocker postpartum given the resolution of cardiovascular changes and risk of maternal bradycardia and hypotension.

THANK YOU!

- **BONUS SLIDES**

- Approach to the Obstetric Patient in a Scenario
 - Pre-pregnancy counselling for women with chronic conditions
- Drugs in Pregnancy
- Normal Labs in Pregnancy
- High Yield Infections in pregnancy
- COVID-19 in Pregnancy
- IBD in pregnancy
- MCQs

- **COVERED ELSEWHERE:**

- Thyroid → ENDO
- Asthma Treatment → RESP
- HIV and others in pregnancy → ID

Unsure about drug safety in pregnancy? Need to counsel a patient on meds planning for pregnancy?

Mothertobaby.org = American resource, good patient summaries under “fact sheets” for meds

BUMPS UK (Best use of Medicine in Pregnancy - <https://www.medicinesinpregnancy.org>)

OB Scenario Approach

Management – Consult and Counsel (“C&C”)

Baby Still In :

- “I will consult my colleagues in Obstetrics to evaluate the health of the baby and placenta” (+/- anesthesia)
- “I will consult Paediatrics” (if at all relevant / peripartum scenario)
- If you are unsure about drugs, mention where you would look it up for safety

Baby is Out:

- Follow-up patient
- Counsel re contraception / breastfeeding + meds
- Counsel re lifetime management of issue
 - GDM → T2DM
 - HTN → Premature Cardiovascular disease, check lipids
- Optimize them before next pregnancy for whatever issue you just dealt with

Counsel this patient on risk of X in pregnancy

1) Counsel them to hold off on conception until disease control (or disease cure)

“The Better Control we can achieve of X prior to conception, the safer your pregnancy will be for you and the more likely that you will have a healthy term baby”

2) Pharmacologic and Non pharmacologic management

→Lifestyle (quit smoking, **QUIT MARIJUANA**, lose weight, exercise will help pretty much everything)

→Advise them on contraception methods that are safe if not getting pregnant

→Pre-conception Folic acid **Folic acid recommendations for NTD prevention (SOGC 2015)**

-Low risk (0.4mg daily, in multivitamin)

-Moderate risk (1mg daily): family history of NTD, maternal type 1 or 2 diabetes, anti-epileptic drugs, GI malabsorption disorders

-High risk (4mg daily): personal history of NTD (maternal or paternal), previous pregnancy with NTD

Review meds for teratogens and change therapy (ex ACEi, valproic acid, warfarin)

3) Consult and Follow

If you get pregnant let me know right away... I will refer you to (MFM, subspecialist with expertise in pregnancy management of X)

Premorbid Condition	MATERNAL / FETAL RISKS
Diabetes (type 1 and 2 similar risks)	FETAL: Stillbirth (4-5X), Perinatal death (3X), Macrosomia/LGA Shoulder dystocia, Birth Trauma, Hypoglycemia MATERNAL: Progression of microvascular disease, Pre-eclampsia
Chronic Hypertension	FETAL: SGA, PreTerm Labour MATERNAL: Pre-eclampsia (*Those with chronic HTN who develop pre-eclampsia have higher perinatal mortality than those who develop 'gestational' HTN → preeclampsia)
Obesity (pre-preg BMI >30)	FETAL: higher rates of some fetal anomalies (Neural Tube), SA, stillbirth MATERNAL: GDM, HTN, OSA, preeclampsia, increased risk of C-Section, VTE
Prior Preeclampsia	1/6 chance repeat Preeclampsia If Prior SEVERE preeclampsia 25% if <34wks, 50% if <28wks MATERNAL: Lifetime CV risk higher (esp. if preeclampsia <34wks GA); Higher lifetime risk of developing diabetes, CKD, VTE.
Autoimmune diseases	General rule of thumb: pre-conception disease control predicts pregnancy outcomes (eg Crohn's in remission = 30% flare; Crohn's active = 70% flare).
Epilepsy	FETAL: Major congenital malformations increased w/ AEDs MATERNAL: no seizures in 9mos prior to pregnancy, 90% remain seizure free; Gestational HTN, Preeclampsia risk higher

Drugs in Pregnancy

- Be Aware that the FDA changed their pregnancy labelling (categories A → X are gone)
 - Uptodate still lists these categories – don't be fooled!
- Vetted by IMR Resources
 - **FREE: Mothertobaby.org** = American resource, great, good patient summaries
 - **FREE: BUMPS UK** (Best use of Medicine in Pregnancy - <https://www.medicinesinpregnancy.org>)
 - **FDA Pregnancy Exposure Registry**
 - Exposure registry (patients register online)
 - Risk Summary
 - Evidence for applicable drugs
 - **Reprotox**
 - Review of medications in pregnancy and lactation, requires subscription
 - **FREE: LACTMED – for breastfeeding – very reliable and free!**
 - Medication in lactation, app/website from NIH
- Know common drugs contraindicated in pregnancy
 - Eg. Accutane, Valproic acid, Methotrexate, Thalidomide, DOACs...
 - Need contraception discussion if prescribing these to woman of childbearing age!

Drug Class	Safe	Avoid
Antibiotics	Macrolides, Penicillins / b-lactam Carbapenems, INH(+B6) Nitrofurantoin (avoid after 36wk – risk of kernicterus)	Fluoroquinolones Tetracyclines Septra avoid esp Tri1 but not C/I
Anticoagulation	LMWH, Unfractionated Hep, ASA	Warfarin (except mech valves), DOACs
Endocrine Meds	Thyroxine Metformin, Insulin, Glyburide Antithyroid drugs (for Graves, no MMI in Tri1)	Non-glyburide Sulfonylureas, TZDs DPP-IV (prob safe but not recommended) SGLT2i, GLP1RAs
GI medications	H2 blockers, PPIs Metoclopramide, Dimenhydrinate, Ondansetron 5-ASA	QT risk
Neurologic Medications	AEDs: lamotrigine and levetiracetam considered safest Pyridostigmine *antidepressants* - risk/benefit favours continue	Some Anti-epileptic drugs* (esp valproate and carbamazepine contraindicated) Benzodiazepines – neonatal withdrawal
Anti-inflammatories	Prednisone, Plaquenil, Cyclosporin, Tacrolimus, Sulfasalazine, Prednisone, IVIG, Colchicine All TNF inhibitors can be used throughout pregnancy	Methotrexate, cyclophosphamide, mycophenolate Tofacitinib – small molecule biologic would X placenta, no safety data
Analgesics	Acetaminophen NSAIDs (use intermittently in first and second trimester, stop after 28 weeks gestation given risk of premature PDA closure). Advil preferred.	Codeine breastfeeding Chronic narcotics – risk of NAS (Methadone and suboxone continue to avoid relapse if addiction)

Vaccines in Pregnancy

NOT IN PREGNANCY:

- Live / Live attenuated vaccines : Varicella, MMR, Rabies, yellow fever, nasal influenza
 - If given pre-pregnancy, counsel to delay getting pregnant 4 wks
- HPV – under study currently

YES IN PREGNANCY:

- COVID mRNA vaccines
- Influenza Inactivated (confers immunity to newborn etc) (not LAIV)
- Tdap in every pregnancy between 21-32 weeks GA (**Ideally between 27-32 wks; SOGC, 2018**)
- **Abrysvo is a bivalent RSV vaccine in pregnancy** [approved Dec 2023] for use in pregnant women between 32-36 weeks GA to reduce risk of severe neonatal RSV infections. Health Canada (NACI) is suggesting Beyfortus (monoclonal Ab given to neonates / infants) is preferred RSV prevention strategy over Abrysvo so Ontario Public Health only paying for one option for mother/child (either Abrysvo or Beyfortus not both).
- Other vaccines if indicated for other reasons
 - eg pneumovax for asthma
 - Hepatitis A or B for travel or household exposure
 - Meningococcal

YES in BREASTFEEDING:

- All Vaccines
SOGC Vaccinations in Pregnancy

Obesity in pregnancy

SOGC Guideline 2019

- Pregnancy risks with maternal obesity:
 - Increased pregnancy loss including miscarriage and stillbirth. Increased odds of operative delivery.
 - Maternal risks: GDM, HTN, Preeclampsia, VTE, UTI
 - Fetal/neonatal Risks: Stillbirth, Shoulder dystocia, Macrosomia, fetal malformations (open neural tube defects, cardiac, spina bifida), hypoglycemia, NICU admission, Preterm birth, meconium aspiration
- US Preventive Services Task Force May 2021 recommendations
 - To offer pregnant persons effective behavioural counseling interventions aimed at promoting healthy weight gain and preventing excess gestational weight gain in pregnancy
 - Behavioural counselling modestly decreased risk of GDM, emergency C/S, macrosomia, and large for gestational size
- Intrapartum
 - Maternal **weight gain targets** from Institute of Medicine **5-10kg for obesity BMI >30**
 - Excessive gestational weight gain associated with pregnancy and fetal complications.
 - Exercise in pregnancy: **target >150min moderate intensity activity / week**
 - **ASA to prevent preeclampsia** if other maternal risks exist
 - Folic acid >1mg/day, Vitamin D >400 IU / day
 - Closer surveillance (OB to discuss with patient, + consider delivery 39-40weeks to reduce risk of stillbirth if BMI >40... details for OB to know, not internist)
- Post-partum
 - Higher risk of PP Depression, anxiety, recommended to screen for same
 - Higher rates of breastfeeding discontinuation – lactation support recommended
 - Thromboprophylaxis if C-Section

Physiologic Changes in Pregnancy: Q

Which of the following is not a physiologic change in pregnancy?

- a) Increase in cardiac output
- b) Peripheral vasodilatation
- c) Decrease in venous return
- d) Decrease in minute ventilation
- e) Hypercoagulable state
- f) Increase in glomerular filtration
- g) Increase in insulin resistance
- h) Decrease in gastrointestinal motility and lower esophageal sphincter tone

Physiologic Changes in Pregnancy:A

Which of the following is not a physiologic change in pregnancy?

- a) Cardiac output and plasma volume increases by 50% in the second trimester and return by 6-8 weeks postpartum
- b) Progesterone is a peripheral vasodilator and decreases peripheral vascular resistance resulting in peripheral edema and orthostatic presyncope/syncope
- c) The gravid uterus compresses on the IVC and pelvic veins leading to decreased venous return, varicose veins, hemorrhoids, and peripheral edema
- d) Progesterone stimulates ventilation causing an increase in tidal volume and minute ventilation, resulting in physiologic dyspnea of pregnancy and respiratory alkalosis**
- e) Pregnancy increases clotting factors and venous stasis causing a hypercoagulable state
- f) Glomerular filtration increases by up to 40% in pregnancy due to increased in blood flow to the kidneys
- g) Placental hormones leads to peripheral insulin resistance giving rise to gestational diabetes
- h) Progesterone decreases gastrointestinal motility and increases transit time causing constipation. It also relaxes the smooth muscle of the lower esophageal sphincter causing acid reflux

Labs in Pregnancy

What is the abnormal lab in a pregnant patient in 3rd trimester?

- a) Hgb 110
- b) WBC 12.9 (8.9 neutrophils)
- c) Plt 124
- d) Cr 96
- e) BG 3.7
- f) ALT 45
- g) ALP 172
- h) Bilirubin 8

Labs in Pregnancy

What is the abnormal lab in a pregnant patient in 3rd trimester?

- a) Hgb 110 – Hb cut-offs in pregnancy: >110 in first trimester, >105 in second and third trimesters, >100 post partum
- b) WBC 12.9 (8.9 neutrophils) – elevated WBC predominantly neutrophils seen in pregnancy due to demargination (like stress response to steroids). WBC up to 13 common and not reflective of infection.
- c) Plt 124 – gestational thrombocytopenia common in third trimester due to hemodilution, usually >100.
- d) **Cr 96 - Cr should drop in pregnancy due to increase in GFR by 40%! Usually Cr of 50 or less in healthy women without renal disease.**
- e) BG 3.7 – normal fasting blood glucose 3.9 +/- 0.4. Due to glucose and amino acids preferentially diverted to the fetus, decrease hepatic gluconeogenesis, and increase in peripheral glucose consumption.
- f) ALT 45
- g) ALP 172 - ALP is made by the placenta, so is normally elevated 2nd, 3rd trimester. AST, ALT, Bilirubin, GGT should be within normal limits in pregnancy.
- h) Bilirubin 8

Infections in Pregnancy

- Special clinical scenarios
 - **Asymptomatic bacteriuria** → Associated w/ PTL, LBW; progression to pyelo
→ TREAT pregnant women with asymptomatic bacteriuria
 - **Genital infections peripartum** → active HSV lesions warrant C-section to reduce neonatal exposure. For women with HSV in pregnancy or hx recurrent disease, consider acyclovir 36+ wks to suppress outbreak.
 - **Listeria** → Suspect if sepsis or meningitis, cover with ampicillin
 - **Syphilis** → even if PCN allergic, best to **desensitize and treat w/ PCN** (alternatives ex. doxy C/I in pregnancy). See ID lecture for dosing.
 - **HIV** → Indication to start ART. PJP can be treated with Septra, ensure high-dose folate. C-section unless VL <1000 and no breastfeeding. Consult Paeds to help w/ Post Exposure Prophylaxis for baby.

COVID-19 in Pregnancy

SOGC 2021-3, CTC guidelines bccdc.ca

- Women who contract COVID 19 in pregnancy, even if mild cases, need to report to OB care provider
 - 3rd trimester growth ultrasound (+/- serial US if concern) following COVID 19 infection
 - Risk of growth restriction, **hypertensive disorders including severe preeclampsia higher following COVID 19 infection**
- Risk of **hospitalization, maternal ICU admission, preterm birth, and infant NICU admission** are **significantly ↓ for those with ≥2 vaccinations compared to the unvaccinated**. (CANCOVID data set, sept 2023)
- **Vaccination is safe** in pregnancy and lactation (follow NACI dosing intervals)
 - no difference in rates of fertility, spontaneous abortion, stillbirth, preterm birth, with vaccination.

SOGC 2022

Treatment of COVID 19 in Pregnancy:

Usual treatment – if sick and admitted to hospital requiring oxygen use standard pharmacotherapies + DVT prophylaxis

Dexamethasone ↓ mortality in those requiring mechanical ventilation or oxygen (Recovery trial), but 90% crosses placenta and can adversely affect fetus. Recovery **included 6 pregnant patients**, given Prednisolone, Hydrocortisone or Methylprednisolone.

- Mount Sinai Hospital consensus
 - <22 weeks or >34 weeks: **Methylprednisolone** 32mg IV for 10 days or discharge [suggested in bccdc.ca]
 - 22-34 weeks gestation: **Dexamethasone** 6mg BID x 2 days (replace Betamethasone for lung maturation) followed by **Methylprednisolone** 32mg IV for 8 days/discharge
 - Postpartum: Dex 6mg daily for 10 days or discharge
- **Paxlovid**, and **Tocilizumab** should be considered in pregnancy if usual adult criteria met
- Remdesivir not favoured due to lack of mortality benefit, less data for safety of baricitinib – case by case. [bccdc.ca]

IBD

(Toronto Consensus statement Gastroenterology 2016; 150: 734-757)

- Pre-pregnancy disease control predicts flares in pregnancy (inc. stillbirth, preterm labour, low birth weight if active disease)
 - Consensus: steroid free **remission x 3 months before conception**
- Pre-conception
 - folate, stop smoking, update vaccines, optimize comorbidities, evaluate disease control.
 - **Discontinue MTX** x 3 months preconception
 - 5-ASA – continue, ensure phthalate free
 - AZATHIOPRINE CONTINUE, anti-TNF continue [strong recommendation for both. Large meta-analysis confirms safety of biologics in pregnancy: O'Byrne LJ et al. BJOG. 2022 Jul;129(8):1236-1246]
- FLARE – Prednisone, 5-ASA, anti-TNF. Mgmt by a **GI specialist**, not a GIM! Also consult **high risk OB**.
 - Consider stress dose at delivery
 - Screen early and frequently for Diabetes
- DELIVERY:
 - Mode (vaginal/c-section) dictated by obstetric reasons
 - Fistulizing perianal Crohn's needs some thought with regard to mode of delivery – defer to GI/OB (C-section likely)
 - Consider VTE prophylaxis postpartum
- Babies exposed to TNF – avoid live vaccines (ex. rotavirus) until >6 mos age

MCQ 1 2025

A 27-year-old G2P1 currently at 31 weeks gestation presents to hospital with blood pressure of 190/120 and HR 95 with shortness of breath. On exam, she is alert, her JVP is 5cm above the sternal angle and she has bilateral crackles. She has a history of chronic hypertension on Labetalol 200mg TID and Nifedipine XL 30mg BID. She has been on Aspirin 162mg nightly since 10 weeks' gestation. Investigations show Hgb 100, Plt 150, ALT 160, Cr 142, urinalysis 3+ protein, UPCR of 140. Fetal ultrasound shows growth in the 8th percentile.

Which of the following is the next best step?

- a) Give Nifedipine 30mg XL STAT and start Methyldopa 500mg TID
- b) Give Hydralazine 10mg IV and nitroglycerin patch, with consideration of Lasix
- c) Give Labetalol 10mg IV and increase Nifedipine XL to 60mg TID
- d) MgSO₄ 4g IV bolus then infusion at 2g/hr
- e) CTPA to assess for evidence of pulmonary edema / dyspnea workup
- f) Outpatient anti-hypertensive management with close follow up

MCQ 2 2025

36F G1P0 currently at 32 weeks gestation presents to a rural hospital with sudden onset pleuritic chest pain and shortness of breath. She denies any unilateral leg, buttock, flank, or back pain. On exam, she is tachycardic in 140s, RR 28, and her oxygen saturation is 90% on room air and 94% on 2L nasal prongs. CXR is unremarkable and labs do not show evidence of pre-eclampsia.

Which of the following is the most appropriate next step?

- a) Treat empirically with Apixaban
- b) Order d-dimer and if <500 PE ruled out
- c) Transfer to a facility with access to a V/Q scan
- d) Start therapeutic LMWH now given high index of suspicion of PE
- e) Order bilateral lower extremity dopplers the following morning the following morning

MCQ 3 2025

A 40F G2P1 planning pregnancy through her local Fertility Clinic. She has a history of Type 2 Diabetes Mellitus on Metformin, Sitagliptin, Gliclazide, and Semaglutide weekly injections. Her BMI is 43 and blood pressure is 135/85. Last HbA1c is 7.8%.

Which of the following statements is *true* in management of pregnant patients with diabetes?

- a) HbA1c should ideally be 7.5% pre-conception
- b) DPP-4 inhibitors, GLP-1 agonists, and SGLT-2 inhibitors are safe in pregnancy and breastfeeding
- c) Metformin and Gliclazide may be continued in pregnancy
- d) She will likely be started on insulin in pregnancy
- e) She should be started on Aspirin 162mg daily now and continued until 36 weeks

MCQ 4 2025

A 29-year-old G1P0 currently at 38 weeks gestation, with a 2-week history of vomiting, polydipsia, and polyuria. On exam, her blood pressure is 145/85, she is alert and oriented, her skin is jaundiced and hyperreflexic. Investigations show Hgb 95, Plt 100, ALT 800, AST 600, ALP 450, GGT 300, bilirubin 80, direct bilirubin 60, INR 2, glucose 2.4, Cr 160. Her abdominal ultrasound showed a normal liver, biliary tract, and negative portal vein dopplers.

Which of the following is the *next* best step?

- a) Start ursodiol 500mg PO BID for cholestasis of pregnancy
- b) Give Nifedipine 10mg IR followed by magnesium sulfate for pre-eclampsia with HELLP
- c) Start Metoclopramide and Gravol for hyperemesis gravidarum
- d) STAT Obstetrics and Hepatology consultation, delivery, supportive management, and liver biopsy for acute fatty liver of pregnancy
- e) Outpatient management of hypertension

MCQ 5 2025

A 32F otherwise healthy Indigenous woman G1P0 at 35 weeks gestational age is seen in clinic with concerns of profound fatigue and reduced exercise tolerance. Her vitals are BP 120/80, HR 120, R 20, afebrile, saturating 98% on room air. Her cardiac and respiratory exam are unremarkable. She indicates she has been chewing more ice than typically. Her blood work indicates a hemoglobin of 97, MCV 75, WBC 6.0, Plt 250. Two months ago, her ferritin was 70, vitamin B12 was 310 .

What is the most appropriate next step?

- a. Repeat blood work and follow up in 2 weeks, as this is a common pregnancy complaint
- b. Repeat ferritin and B12, order hemoglobin electrophoresis
- c. Repeat CBC, ferritin, B12 and arrange for iron sucrose infusion 300mg x 1
- d. Admit to hospital and order a hemolytic work up

MCQ 6A 2025

A 28F with a history of chronic hypertension G2P1 at 32 weeks gestational age is seen in clinic with hand tingling. Her pregnancy thus far has been uncomplicated, she is on labetalol 200mg PO BID. Her Oral Glucose Test was within normal limits. She indicates this is worse after her desk job and at night. Her vitals are BP 130/80, HR 95, R 18, afebrile, saturating 97% on room air. Her neurologic examination with within normal limits including normal reflexes and no clonus. Her cardiac and respiratory exam are unremarkable. Her hands are mildly swollen and positive Tinel's sign. She does not have abdominal pain, facial or lower extremity edema. Her most recent blood work is within normal limits.

What is the most likely diagnosis?

- a. Pre-eclampsia
- b. Gestational diabetes
- c. Carpal tunnel syndrome
- d. Radial nerve palsy
- e. Neuralgic amyotrophy

MCQ 6B 2025

A 28F with a history of chronic hypertension G2P1 at 32 weeks gestational age is seen in clinic with hand tingling. Her pregnancy thus far has been uncomplicated, she is on labetalol 200mg PO BID. Her Oral Glucose Test was within normal limits. She indicates this is worse after her desk job and at night. Her vitals are BP 130/80, HR 95, R 18, afebrile, saturating 97% on room air. Her neurologic examination with within normal limits including normal reflexes and no clonus. Her cardiac and respiratory exam are unremarkable. Her hands are mildly swollen and positive Tinel's sign. She does not have abdominal pain, facial or lower extremity edema. Her most recent blood work is within normal limits.

What is the next best step in management?

- a. Admit to hospital for blood work (CBC, electrolytes, Cr, Liver enzymes, UPCR) and monitoring
- b. Refer to Neurology for EMG studies
- c. Prescribe bilateral wrist splints and reassure her this can occur in pregnancy due to fluid changes
- d. Repeat A1c, blood glucose and lipid panel
- e. Decompressive surgery

MCQ 7 2025

A 40F otherwise healthy G2P1 at 28 weeks gestational age is referred to you for palpitations. Her vitals are: BP 115/75, HR 125, R 18, saturating 98% on room air. Her cardiac and respiratory examination are within normal limits. She does not have a tremor. ECG shows sinus tachycardia, her 48-hour Holter monitor shows 10% PVCs and no other arrhythmias.

Which of the follow is not recommended?

- a. Obtain electrolytes, extended electrolytes, ferritin, TSH
- b. Start amiodarone for symptomatic PVCs
- c. Start metoprolol for symptomatic PVCs
- d. Arrange for outpatient TTE if her symptoms persist
- e. Monitor for symptoms of heart failure

MCQ 8 2025

35F with GDM. Strictly adherent to recommended diet on review of her diet log and is exercising 30 min/day (walking). From her blood glucose log for the 7 days shows her 1 hr post prandial sugars:

Post breakfast 8-8.2

Post lunch 7.8-7.9

Post dinner 7.2-7.3. and AM fasting 5.1-5.3.

What insulin regimen would you start?

- a. 2 units of aspart with breakfast
- b. 2 units of aspart with breakfast and lunch
- c. 2 units of aspart with breakfast and lunch and 2 units of NPH at bedtime
- d. Glargine 8 u at bedtime

MCQ 9 2025

A 24y G1P1 woman experiences preeclampsia at 36 weeks and her daughter is delivered. She is seen 6 weeks postpartum. She is healthy with normal BP, BMI and has no history of diabetes. She is presently breastfeeding.

In follow-up which is false?

- a) She should be counselled that breastfeeding reduces risk of future cardiovascular diseases.
- b) In future pregnancies she should take ASA 162mg to prevent preeclampsia from 12-36 weeks.
- c) A lipid profile should be drawn at nearest opportunity and statin therapy initiated if intermediate-high risk
- d) Moderate intensity Exercise 150 min /week is recommended in pregnancy to prevent risk of preeclampsia

MCQ 10 2025

32F G2P1 currently 8 weeks gestation has a history of estrogen containing oral contraceptive related DVT five years ago. She is currently asymptomatic and weighs 70kg.

Which of the following is the *least* appropriate?

- a) Her risk of VTE recurrence is 5-10% in pregnancy
- b) Start Enoxaparin 40mg sc daily now until 6 weeks post partum
- c) Stop Enoxaparin at least 12 hours before an epidural peripartum
- d) Given maternal weight gain in pregnancy, anti-Xa levels should be monitored
- e) She should be switched to therapeutic Enoxaparin if diagnosed with VTE in pregnancy

MCQ 11 2025

A 42 year old G3P3 with type 2 diabetes is planning a pregnancy. Her HbA1c is 7% on Metformin, Sitagliptin and Semaglutide. She is also on Atorvastatin and Perindopril. Her blood pressure is 125/75, Cr 85 with ACR of 35.

Which of the following statements is false?

- a) She may require insulin during pregnancy
- b) Continue Metformin, discontinue Sitagliptin and Semaglutide
- c) Continue Perindopril until positive pregnancy test
- d) Continue Atorvastatin until positive pregnancy test
- e) Target BP <130/80 during pregnancy
- f) She is at increased risk of pre-eclampsia and will require ASA 162mg at 10-12 weeks gestational age

BONUS MCQ Answers

MCQ 1: B - Hypertensive emergency in pregnancy: Use rapid acting anti-hypertensives (eg Hydralazine, Labetalol) paired with long-acting med and r/o pre-eclampsia. Given CHF favour hydralazine over labetalol (beta blocking effect).

MCQ 2: D - This is a scenario where PE is high likely – re Di Dimer statement is referring to the YEARS algorithm which should really only be applied in low to intermediate risk pre-test probability patients and is not the most appropriate next step

MCQ 3: D - Correct, given metformin is the only medication in her current regimen she can continue she will need better glycemic control

MCQ 4: D - Liver disorder management in pregnancy: This is a case of acute fatty liver of pregnancy, seen late in pregnancy where microvesicular fatty infiltration secondary to mitochondrial defect in fatty acid metabolism (LCHAD) in fetus leads to maternal liver failure. This patient has more than 6 of 14 Swansea Criteria for diagnosis. Treatment is delivery and supportive management with prompt diagnosis and monitoring of liver function post-delivery.

BONUS MCQ Answers

MCQ 5: C – repeat CBC, ferritin, B12 and give IV iron sucrose. Given the progression of her pregnancy, microcytic anemia with previously normal ferritin 2 months ago, this is likely iron deficiency anemia. She is close to term and should she go into spontaneous labor, would benefit from IV iron now. Hemoglobin electrophoresis should be considered in at risk ethnicity- Indigenous incl First Nations/Inuit, Northern European Caucasians, Asians of Japanese and Korean descent are NOT at risk of thalassemia (SOGC guideline).

MCQ 6A: C – Carpal tunnel syndrome. As pregnancy progresses, fluid retention increases and presenting symptoms of carpal tunnel syndrome are similar to those nonpregnant patients (numbness, paresthesia along median nerve, wrist pain, night awakening, etc). Diabetes, Hypothyroidism, Pregnancy all risk factors

MCQ 6B: C – Prescribe bilateral wrist splints and reassure. This not uncommon in pregnancy, gather a detailed history, perform a physical exam, and treatment is activity medication, wrist splinting in a neutral position. Often resolves a few weeks after childbirth and can be correlated with weight gained (fluid retention) during pregnancy

MCQ 7: B – Avoid Amiodarone in pregnancy. In addition to checking metabolic labs you would also get an ECHO and refer to a cardiologist – 10% is a high burden of PVCs even if not pregnant in a young woman.

BONUS MCQ Answers

MCQ 8: B - recall GDM targets ≤ 5.3 fasting, and ≤ 7.8 1 h post meal. She is on target with her fasting glucose and her post-dinner. She needs prandial **insulin with breakfast and lunch**. An alternative would be to try metformin in daytime but this is not listed as an option.

MCQ 9: C – drawing lipid profile and starting statin. JAMA 2022 meta-analysis of >1million parous women showed reduction in MI, Stroke, CV death in women who breastfed controlling for multivariable. Prior preeclampsia is an indication for ASA. In all women with hypertensive disorders of pregnancy CCS suggests lipid profile be drawn postpartum but frankly we aren't given a lot of guidance as to what to do with results in a 24-year-old – it's to inform decision making. Certainly, you won't start a statin in a breastfeeding woman. Exercise is in SOGC guideline to reduce risk of preeclampsia (as long as no contraindication eg placenta previa)

MCQ 10: D – measure anti-Xa levels. Do not measure anti-Xa levels on prophylactic LMWH. Xa levels can be considered for therapeutic anticoagulation for extremes of weight, breakthrough VTE, renal dysfunction.

MCQ 11: D – continue atorvastatin until positive pregnancy test. Should stop statins 3 months pre-pregnancy or while attempting to conceive due to uncertainty regarding teratogenic effects or miscarriage risk, in this scenario for primary prevention so can stop. ACEi are not recommended in pregnancy but harm occurs in Tri2/3 (renal agenesis, oligohydramnios due to same), when indicated for proteinuria some continue until positive preg test.

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